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0.1 Definition [Notation]

Let $x, y \in \mathbb{R}^d, \alpha = (\alpha_1, \dots, \alpha_d) \in \mathbb{N}^d$

$$x \cdot y = (x | y)_{\ell^2} = \sum_{j=1}^d x_j y_j, \quad |x| = \|x\|_{\ell^2}$$

$$x^\alpha = x_1^{\alpha_1} \dots x_d^{\alpha_d}, \quad \langle x \rangle = (1 + |x|^2)^{\frac{1}{2}}$$

$$(f | g) = \int f g$$

$$(f(x, y) | g(x, y))_x = \int f(t, y) g(t, y) dt$$

$$(f(x) | g(x))_{L^2} = \int f(x) \bar{g}(x) dx$$

$$\langle T | \varphi \rangle_{X^*, X} = T(\varphi)$$

0.2 Definition [Modified Vinogradov Notation]

Two quantities are related as

$$A \lesssim B$$

if there exists c independent of A, B ; such that $A \leq cB$.

$A \sim B$ means $A \lesssim B$ and $B \lesssim A$

0.3 Definition [Landau Symbols]

Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$ be functions and let $x_0 \in \mathbb{R} \cup \{\pm\infty\}$ with $g(x) \neq 0$ in a punctured neighborhood of x_0 .

1) $f(x) = o(g(x))$ as $x \rightarrow x_0$ if

$$\lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = 0.$$

2) $f(x) = \mathcal{O}(g(x))$ as $x \rightarrow x_0$ if there exist constants $C > 0$ and $\delta > 0$ such that

$$|f(x)| \leq C |g(x)| \quad \text{for all } x \text{ with } 0 < |x - x_0| < \delta.$$

Alternatively,

$$|f(x)| \lesssim |g(x)|, \quad x \in B(x_0, \delta)$$

1.0 Functional Analysis

1.1 Measure Theory

1.1 Definition [Push-Forward Measure]

Let $(X, \mathcal{A}, \mu)$ be a measure space. Let $(Y, \mathcal{G})$ be a measurable space and

$$T : X \rightarrow Y$$

be a measurable map. Then T induces a measure on $(Y, \mathcal{G})$ from μ .

$$\mu_T(B) := \mu(T^{-1}(B)), \quad \forall B \in \mathcal{G}$$

There is also a notation $\mu_T = T_{\#}\mu$

1.2 Definition [Absolute Continuity]

Let $(X, \mathcal{A})$ be a measurable space and μ, ν be two measures on it. We say that μ is absolutely continuous w.r.t. ν

$$\mu \ll \nu$$

iff

$$\nu(A) = 0 \implies \mu(A) = 0, \quad \forall A \in \mathcal{A}$$

Meaning that null sets of ν must also be null sets of μ .

μ can still have other null sets, thus μ contains less information than ν .

1.3 Theorem [Radon-Nikodym]

Let $(X, \mathcal{A})$ be a measurable space and μ, ν be two measures on it with

$$\mu \ll \nu$$

Then, there exists a measurable $f : X \rightarrow [0, \infty)$ such that

$$\int_A d\mu = \int_A f d\nu$$

Also written

$$d\mu = f d\nu,$$

motivating the terminology f : Radon-Nikodym derivative of μ w.r.t. ν

$$f = \frac{d\mu}{d\nu}$$

In probability, with $\nu = \text{Leb}_{\mathbb{R}}$, such f is usually called pdf (probability density function). Intuitively, we rescale ν with f in some way to get the same measure. The absolute continuity is necessary because we can scale down to zero by f but can't retrieve back the information.

1.4 Theorem [Transfer Theorem]

Let $(X, \mathcal{A}, \mu), (Y, \mathcal{G}, \mu_T), (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ be measure spaces. Let

$$T : (X, \mathcal{A}) \rightarrow (Y, \mathcal{G}), \quad g : (Y, \mathcal{G}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$$

Then, $g(T) \in L^1(X, \mathcal{A}, \mu)$ iff $g \in L^1(Y, \mathcal{G}, \mu_T)$, and then we have

$$\int_X g(T(x)) d\mu(x) = \int_Y g(y) d\mu_T(y)$$

Usually $Y = \mathbb{R}^d, \mathcal{G} = \mathcal{B}(\mathbb{R}^d)$

(Doesn't require Radon-Nikodym)

1.2 Integrals

Main Integral Theorems

1.5 Theorem [(Fubini-)Tonelli]

Let $(X, \mathcal{A}, \mu)$ and $(Y, \mathcal{B}, \nu)$ be σ -finite measure spaces. Let $f : X \times Y \rightarrow [0, \infty]$ be $\mu \otimes \nu$ measurable.

Then,

$$\int_{X \times Y} f d(\mu \otimes \nu) = \int_X \left[\int_Y f(x, y) d\nu(y) \right] d\mu(x) = \int_Y \left[\int_X f(x, y) d\mu(x) \right] d\nu(y).$$

1.6 Theorem [Fubini]

Let $(X, \mathcal{A}, \mu)$ and $(Y, \mathcal{B}, \nu)$ be σ -finite measure spaces, and let $f : X \times Y \rightarrow \mathbb{R}$ be $L^1(X \times Y, \mathcal{A} \otimes \mathcal{B}, \mu \otimes \nu)$, i.e.

$$\int_{X \times Y} |f(x, y)| \, d(\mu \otimes \nu)(x, y) < \infty.$$

Then,

$$\int_{X \times Y} f \, d(\mu \otimes \nu) = \int_X \left[\int_Y f(x, y) \, d\nu(y) \right] d\mu(x) = \int_Y \left[\int_X f(x, y) \, d\mu(x) \right] d\nu(y),$$

and the iterated integrals exist and are finite.

1.7 Theorem [Dominated Convergence]

Let $(f_n)_{n \in \mathbb{N}}$ be a sequence of measurable functions on $(X, \mathcal{A}, \mu)$, such that

i) $\exists f : X \rightarrow \mathbb{R}$ μ -measurable with

$$f_n \xrightarrow[n \rightarrow \infty]{a.e.} f$$

ii) $\exists g \in L^1(X, \mathcal{A}, \mu)$ with

$$g \geq 0, \quad |f_n| \leq g, \forall n$$

Then, $f_n, f \in L^1(X, \mathcal{A}, \mu)$ with

$$\int_X |f_n - f| \, d\mu \xrightarrow[n \rightarrow \infty]{} 0$$

which implies

$$\lim_{n \rightarrow \infty} \int_X f_n \, d\mu = \int_X \lim_{n \rightarrow \infty} f_n \, d\mu = \int_X f \, d\mu$$

1.8 Theorem [Monotone Convergence]

Let $(f_n)_{n \in \mathbb{N}}$ be a sequence of measurable functions on $(X, \mathcal{A}, \mu)$, such that

i) $f_n(x) \nearrow f(x)$ almost everywhere on X .

Then, $f : X \rightarrow [0, \infty]$ is μ -measurable and

$$\lim_{n \rightarrow \infty} \int_X f_n \, d\mu = \int_X \lim_{n \rightarrow \infty} f_n \, d\mu = \int_X f \, d\mu.$$

1.9 Theorem [Fatou's Lemma]

Let $(f_n)_{n \in \mathbb{N}}$ be a sequence of measurable functions on $(X, \mathcal{A}, \mu)$.

Define $f(x) := \liminf_{n \rightarrow \infty} f_n(x)$, then $f : X \rightarrow [0, \infty]$ is measurable. **Then**,

$$\int_X f \, d\mu = \int_X \liminf_{n \rightarrow \infty} f_n \, d\mu \leq \liminf_{n \rightarrow \infty} \int_X f_n \, d\mu.$$

Integrals depending on a parameter

1.10 Theorem [Continuity Under the Integral Sign]

Let (T, d) be a metric space (of parameters), $t^* \in T$, and $f : T \times X \rightarrow \mathbb{C}$ with

- i) for every $t \in T$ fixed, $x \mapsto f(t, x) \in L^1(X, \mathcal{A}, \mu)$
- ii) for a.e. $x \in X$ fixed, $t \mapsto f(t, x)$ is continuous at t^*
- iii) $\exists g : X \rightarrow \mathbb{R}_+$ in $L^1(X, \mathcal{A}, \mu)$, such that for every $t \in T$

$$|f(t, x)| \leq g(x), \quad \text{for a.e. } x$$

Then,

$$F(t) := \int_X f(t, x) \, d\mu(x)$$

is continuous at t^*

1.11 Theorem [Differentiability Under the Integral Sign]

Let $I \subset \mathbb{R}$ be an interval, and $f : I \times X \rightarrow \mathbb{C}$ with

- i) for every $t \in I$ fixed, $x \mapsto f(t, x) \in L^1(X, \mathcal{A}, \mu)$

- ii) for a.e. $x \in X$ fixed, $t \mapsto f(t, x)$ is differentiable on I
- iii) $\exists g : X \rightarrow \mathbb{R}_+$ in $L^1(X, \mathcal{A}, \mu)$, such that for every $t \in T$

$$|\partial_t f(t, x)| \leq g(x), \quad \text{for a.e. } x$$

Then,

$$F(t) := \int_X f(t, x) \, d\mu(x)$$

is differentiable on I , and for all $t \in I$, its derivative is

$$\partial_t F(t) = \int_X \partial_t f(t, x) \, d\mu(x)$$

1.3 Continuous Linear Operators

1.12 Theorem [Vector Spaces]

Let $(E, \| \cdot \|, \mathbb{K})$ be a finite dimensional vector space.

- all norms on E are equivalent
- $(E, \| \cdot \|)$ is Banach
- closed and bounded is equivalent to compact

Let E be a vector space.

- any finite dimensional subset of E is closed
- $\text{span}(x_1, x_2, \dots)$ is closed if it spans a finite dimensional subspace, otherwise, it might be open

1.13 Theorem [Characterization of Continuous Linear Maps]

Let E, F be normed, $T : E \rightarrow F$ be a linear map. The following are equivalent

- 1) T is continuous at 0
- 2) T is continuous on E

3) T is uniformly continuous on E

4) T is Lipschitz on E

5) T is bounded in the following sense

$$(\exists M > 0 : \forall x \in E) : \|T(x)\|_F \leq M \|x\|_E$$

The space of all continuous linear operators between E, F will be denoted $\mathcal{L}(E, F)$

1.14 Definition

$\mathcal{L}(E, F)$ can be equipped with

$$\|T\|_{\mathcal{L}(E, F)} = \sup_{\|x\|_E=1} \|T(x)\|_F$$

The operator norm will be denoted either by $\|\cdot\|_{\mathcal{L}(E, F)}$ or $\|\cdot\|_{E \rightarrow F}$

1.15 Theorem

Let F Banach, then $(\mathcal{L}(E, F), \|\cdot\|_{\mathcal{L}(E, F)})$ is Banach.

1.16 Proposition

Let E be a finite dimensional normed vector space, F normed. **Then**, any linear map $T : E \rightarrow F$ is continuous.

1.17 Proposition

Let $A \in \mathcal{L}(D, E), B \in \mathcal{L}(E, F)$, then, $B \circ A \in \mathcal{L}(D, F)$ with

$$\|B \circ A\|_{\mathcal{L}(D, F)} \leq \|A\|_{\mathcal{L}(D, E)} \|B\|_{\mathcal{L}(E, F)}$$

1.18 Theorem [Continuous Linear Extension]

Let E be normed, F Banach, D dense in E . Let $T_0 \in \mathcal{L}(D, F)$. **Then**, there exists

the unique extension $T \in \mathcal{L}(E, F)$ with

$$T|_D = T_0, \quad \|T\|_{\mathcal{L}(E, F)} = \|T_0\|_{\mathcal{L}(D, F)}$$

1.4 Hahn-Banach Theorems

1.19 Theorem [Hahn-Banach Real Analytic Form]

Let E be a vector space over $\mathbb{R}$. Let $G \leq E$ be a **linear subspace**, $p : E \rightarrow \mathbb{R}$ be a **sublinear function**

$$\begin{aligned} p(\lambda x) &= \lambda p(x), & \forall x \in E, \forall \lambda > 0 \\ p(x + y) &\leq p(x) + p(y), & \forall x, y \in E \end{aligned}$$

Let $\ell_0 : G \rightarrow \mathbb{R}$ be a **linear functional** with

$$\ell_0(x) \leq p(x), \forall x \in G$$

Then, there exists a **linear functional** $\ell : E \rightarrow \mathbb{R}$ with

$$\ell|_G = \ell_0, \quad \ell \leq p$$

Proved by Zorn's lemma, non-constructive.

1.20 Theorem [Hahn-Banach for Normed Spaces]

Let $(E, \|\cdot\|)$ be a normed vector space over $\mathbb{K}$. Let $G \leq E$ be a **linear subspace**, $\ell_0 \in G^*$. **Then**, there exists a **linear functional** $\ell \in (E^*, \|\cdot\|_{E^*})$ with

$$\ell|_G = \ell_0, \quad \|\ell\|_{E^*} = \|\ell_0\|_{G^*}$$

1.21 Theorem [Hahn-Banach for Vanishing Forms]

Let $(E, \|\cdot\|)$ be a normed vector space over $\mathbb{K}$. Let $G < E$ be a **proper linear**

subspace. Then, there exists a linear functional $\ell \in (E^, \|\cdot\|_{E^*})$ with*

$$\ell|_G \equiv 0, \quad \ell \not\equiv 0$$

1.22 Theorem [Hahn-Banach Geometric Form]

*Let $A, B \subset E$ be two disjoint, non-empty, subsets. Let one of them be open. Then, there exists a **closed hyperplane** that separates A and B .*

Where a hyperplane is an affine subspace of codimension 1

$$H = [\ell = \alpha] = \{x \in E ; \ell(x) = \alpha\}$$

is closed iff $\ell : E \rightarrow \mathbb{R}$ is continuous.

1.23 Theorem [Hahn-Banach 2nd Geometric Form]

Let $A \subset E, B \subset E$ be two disjoint, non-empty, subsets. Let A be closed and B compact. Then, there exists a closed hyperplane that strictly separates A and B .

1.5 Duality

1.24 Definition [Linear Forms, Dual Space]

Let $(E, \mathbb{K}, \|\cdot\|)$ be a normed space, then $\mathcal{L}(E, \mathbb{K}) = E^*$, a space of continuous linear forms, is called a dual space.

1.25 Definition [Weak and Weak* Topologies]

Let E be a normed space and E^* its dual.

The **weak topology** $\sigma(E, E^*)$ on E is the coarsest topology such that all (continuous in strong norm topology) functionals $\ell \in E^*$ remain continuous.

We say that $(x_n) \subset E$ converges weakly to $x \in E$

$$x_n \xrightarrow{E} x$$

iff

$$\langle \ell | x_n \rangle_{E^*, E} \xrightarrow[n \rightarrow \infty]{} \langle \ell | x \rangle_{E^*, E} \quad (\forall \ell \in E^*)$$

The **weak* topology** $\sigma(E^*, E)$ on E^* is the coarsest topology such that all evaluation maps

$$x \mapsto \ell(x) = \langle \ell | x \rangle, \quad x \in E,$$

remain continuous.

We say that $(\ell_n) \subset E^*$ converges weak* to $\ell \in E^*$

$$\ell_n \xrightarrow{E^*} \ell$$

iff

$$\langle \ell_n | x \rangle_{E^*, E} \xrightarrow[n \rightarrow \infty]{} \langle \ell | x \rangle_{E^*, E} \quad (\forall x \in E)$$

1.26 Theorem [Dual of Banach]

Let E be Banach, then E^* is Banach.

1.27 Definition [Reflexivity]

Let E be normed and E^* be its dual. Let $E^{**} = (E^*)^*$, and denote the canonical embedding

$$J : E \rightarrow E^{**}, \quad J_\ell(x) = \langle \ell | x \rangle_{E^*, E}$$

J is a (linear) isometric injection.

If J is also a surjection, then E is called a reflexive space.

1.28 Definition [Transpose]

Let E be a normed space, $L \in \mathcal{L}(E, E)$. Then, we can define the transpose $L^\top \in \mathcal{L}(E^*, E^*)$ by the relation

$$\langle L^\top \ell | x \rangle_{E^*, E} := \langle \ell | Lx \rangle_{E^*, E}$$

If $L = L^\top$, an operator is called self-transpose.

1.29 Theorem [Fencel–Moreau]

Assume that $\varphi : E \rightarrow (-\infty, \infty]$ is convex, l.s.c., and $\varphi \not\equiv \infty$. **Then**, $\varphi^{**} = \varphi$.
Where $\varphi^* : E^* \rightarrow (-\infty, \infty]$, a Legendre–Fenchel transform of φ , is given by

$$\varphi^*(f) = \sup_{x \in E} \{\langle f | x \rangle - \varphi(x)\}, \quad f \in E^*$$

1.6 Banach Spaces

1.30 Theorem [Baire]

Let X be a complete metric space and let $(X_n)_{n \in \mathbb{N}}$ be a sequence of closed subsets in X with

$$\text{Int}(X_n) = \emptyset, \quad \forall n \in \mathbb{N}$$

Then,

$$\text{Int} \left(\bigcup_{n \in \mathbb{N}} X_n \right) = \emptyset$$

1.31 Theorem [Characterization via Absolute Convergence]

Let $(X, \| \cdot \|)$ be a normed vector space. The following are equivalent:

- i) X is a Banach
- ii) Every absolutely convergent series in X is convergent, i.e.,

$$\sum_{n=1}^{\infty} \|x_n\| < \infty \implies \sum_{n=1}^{\infty} x_n \text{ converges in } X.$$

1.32 Theorem [Banach–Steinhaus Uniform Boundedness Principle]

Let E, F be Banach and let $(T_i)_{i \in I}$ be a family in $\mathcal{L}(E, F)$ with

$$\sup_{i \in I} \|T_i x\|_F < \infty, \quad \forall x \in E.$$

Then,

$$\sup_{i \in I} \|T_i\|_{\mathcal{L}(E, F)} < \infty$$

Alternatively, there exists c

$$\|T_i x\| \leq c \|x\|, \quad \forall x \in E, \forall i \in I.$$

1.33 Theorem [Open Mapping]

Let E, F be Banach and let $T \in \mathcal{L}(E, F)$ surjective. **Then**, there exists $c > 0$, such that

$$T(B_E(0, 1)) \supset B_F(0, c).$$

In other words, T is an open mapping.

1.34 Theorem [Closed Graph]

Let E, F be Banach and let T be linear. If the graph $\Gamma(T)$ is closed in $E \times F$, **then**, T is continuous.

1.35 Theorem [Kakutani]

Let E be Banach. E is reflexive **iff** $\overline{B_E}$ is compact in the weak topology $\sigma(E, E^*)$

1.36 Theorem [Riesz]

Let $(E, \|\cdot\|)$ be normed. $\overline{B_E}$ is compact **iff** E is finite dimensional.

1.37 Theorem [Ascoli–Arzelà]

Let K be a compact metric space and let $\mathcal{H}$ be a bounded subset of $C(K; K)$. If $\mathcal{H}$ is uniformly equicontinuous, that is,

$$\left(\forall f \in \mathcal{H} : \forall \varepsilon > 0 : \exists \delta > 0 \right) \quad d(x, y) < \delta \implies |f(x) - f(y)| < \varepsilon$$

Then, the closure of $\mathcal{H}$ in $C(K; K)$ is compact

1.38 Theorem [Banach–Alaoglu]

Let E be normed. Then, the closed unit ball $\overline{B_{E^*}}$ is compact in E^* , w.r.t. the weak* topology $\sigma(E^*, E)$

1.39 Theorem [Banach–Picard Contractive Mapping Theorem]

Contractive functions on complete metric spaces uniquely fix a point.

Let (X, d) be a non-empty complete metric space, $K < 1$,

$f : X \rightarrow X$ be a continuous mapping with property

$$d(f(x), f(y)) \leq Kd(x, y), \quad \forall x, y \in X.$$

Then, there exists a unique x_* , such that $f(x_*) = x_*$

Such f is called a **strict contraction** and x_* is called a **fixed-point**.

1.7 L^p Spaces

1.40 Theorem [Fischer–Riesz]

L^p spaces are Banach for $p \in [1, \infty]$

1.41 Theorem [Kolmogorov–M. Riesz–Fréchet]

Let $\mathcal{F} \subset L^p(\mathbb{R}^d)$ be bounded, with $p \in [1, \infty)$. If

$$\lim_{|h| \rightarrow 0} \|\tau_h f - f\|_{L^p(\mathbb{R}^d)} = 0, \quad \text{uniformly in } f \in \mathcal{F}$$

then, the closure of restriction $\overline{\mathcal{F}|_{\Omega}}^{\|\cdot\|_{L^p(\Omega)}}$ is compact for any measurable $\Omega \subset \mathbb{R}^d$

1.42 Proposition [Reflexivity and Separability of L^p spaces]

For $p \in (1, \infty)$, the space $L^p(\Omega)$ is reflexive and separable. For L^1 is separable but not reflexive. L^∞ is neither separable nor reflexive.

1.43 Theorem [Duality of L^p]

1.44 Definition [Convolution]

1.45 Definition [Good Kernels]

A sequence $(\delta_n)_n \subset L^1(\mathbb{R}^d)$ is called a **family of good kernels** or an **approximation of identity** iff

1.8 Hilbert Spaces

Geometry, Orthogonality, Projections

1.46 Theorem [Cauchy–Schwarz Inequality]

Let $(H, (\cdot | \cdot))$ be an inner product space. Then, $\forall x, y \in H$

$$| (x | y) | \leq \|x\| \|y\| = \sqrt{(x | x)} \sqrt{(y | y)}$$

1.47 Definition [Orthogonality]

Let $(H, (\cdot | \cdot))$ be an inner product space.

- $x, y \in H$ are called **orthogonal** iff $(x | y) = 0$
- $A \subset H$, $A^\perp = \{y \in H : \forall x \in A ; (x | y) = 0\}$

1.48 Theorem [Parallelogram Identity]

Let $(H, (\cdot | \cdot))$ be an inner product space. Then, $\forall x, y \in H$,

$$\|x + y\|^2 + \|x - y\|^2 = 2\|x\|^2 + 2\|y\|^2.$$

1.49 Theorem [Polarization Identity]

Let $(H, (\cdot | \cdot))$ be an inner product space over $\mathbb{R}$ or $\mathbb{C}$. For all $x, y \in H$, the inner

product can be recovered from the norm by

$$(x | y) = \begin{cases} \frac{\|x + y\|^2 - \|x - y\|^2}{4}, & \text{if } H \text{ is real} \\ \frac{\|x + y\|^2 - \|x - y\|^2 + i\|x + iy\|^2 - i\|x - iy\|^2}{4}, & \text{if } H \text{ is complex} \end{cases}$$

1.50 Theorem [Pythagoras Identity]

Let $(H, (|))$ be an inner product space. Then for all orthogonal systems $x_1, \dots, x_n$

$$\|x_1 + \dots + x_n\|^2 = \|x_1\|^2 + \dots + \|x_n\|^2$$

1.51 Theorem [Fréchet–von Neumann–Jordan]

Let $(E, \| \cdot \|)$ be normed. This norm is Hilbert norm, i.e. rising from some inner product, *iff* it satisfies the parallelogram identity.

Moreover, in this case, it is explicitly derived by polarization identity.

1.52 Theorem [Closed Subspaces, Characterization]

1.53 Proposition

Let $K \leq H$, then K is dense in H iff $K^\perp = \{0\}$

Hilbert Space Theorems

1.54 Theorem [Riesz—Fréchet Representation]

Let H Hilbert. Given any $\varphi \in H^*$, there exists a unique $f_\varphi \in H$, such that

$$\langle \varphi | u \rangle_{H^*, H} = \varphi(u) = (f_\varphi | u)_H, \quad \forall u \in H$$

Moreover,

$$\|f_\varphi\|_H = \|\varphi\|_{H^*}$$

1.55 Theorem [Stampacchia]

Let $a(u, v)$ be a continuous coercive bilinear form on H Hilbert. Let $K \subset H$ be a non-empty closed and convex subset. **Then**, given any $\varphi \in H^*$, there exists a unique $u \in K$, such that

$$\varphi(v - u) \leq a(u, v - u), \quad \forall v \in K$$

1.56 Theorem [Lax–Milgram]

Let $a(u, v)$ be a *continuous coercive (elliptic) bilinear form* on H Hilbert.

$$|a(u, v)| \leq C \|u\|_H \|v\|_H, \quad a(v, v) \geq \alpha \|v\|_H^2$$

Then, given any $\varphi \in H^*$, there exists a unique $u \in H$, such that

$$a(u, v) = \ell(v), \quad \forall v \in H.$$

and it holds that

$$\|u\|_H \leq \frac{1}{\alpha} \|\ell\|_{H^*}$$

Moreover, if a is *symmetric*, then u is characterized by

$$u = \arg \min_{\ell \in H} \left\{ \frac{1}{2} a(v, v) - \ell(v) \right\}$$

1.57 Theorem [Lax–Milgram Operator Version]

Equivalent form for $a(u, v) = (Au | v)_H$ on H Hilbert.

Let $A : H \rightarrow H$ be a linear operator with

$$|(Au | v)_H| \leq C \|u\|_H \|v\|_H, \quad (Av | v)_H \geq \alpha \|v\|_H^2$$

Then, A is a bijection with a bounded inverse A^{-1} , satisfying

$$\|A^{-1}\| \leq \frac{1}{\alpha}$$

Self-Adjoint Operators

1.58 Definition [Adjoint]

Let H Hilbert, $A \in \mathcal{L}(H)$, its adjoint A^* is defined via

$$(A^*u | v)_H = (u | Av)_H$$

If $A = A^*$, an operator is called self-adjoint.

Hilbert Bases

1.9 Distributions

1.59 Definition [Bump Functions C_c^∞]

Let $\Omega \subseteq \mathbb{R}^d$ be open. The space of smooth functions $f : \Omega \rightarrow \mathbb{C}$ with compact support (bump functions / test functions) is denoted by $\mathcal{D}(\Omega) = C_c^\infty(\Omega; \mathbb{C})$ or $C_0^\infty(\Omega; \mathbb{C})$. One can also choose $C_c^\infty(\Omega; \mathbb{R})$.

1.60 Proposition [Density of Bump Functions]

1.61 Definition [Distributions]

Let $\Omega \subseteq \mathbb{R}^d$ be open. The space of **distributions** is the topological dual of the space of test functions:

$$\mathcal{D}^*(\Omega) = (\mathcal{D}(\Omega))^*$$

where $\mathcal{D}^*(\Omega)$ consists of all continuous linear functionals (distributions)

$$T : \mathcal{D}(\Omega) \rightarrow \mathbb{C}$$

Where continuity is with respect to the strong LF topology, from which we can derive the weak^{*} topology $\sigma(\mathcal{D}^*, \mathcal{D})$.

Distributions, also called **generalized functions**, will be then defined not pointwise, like functions, but w.r.t. to averages, given by the integrals against test functions. If we think of f as a distribution T_f ,

1.62 Example

Let $f \in L^1_{\text{loc}}(\Omega)$. Then f defines a distribution T_f as

$$T_f(\varphi) = \langle T_f | \varphi \rangle := (T_f | \varphi) = \int_{\Omega} f(x) \varphi(x) \, dx$$

1.63 Example

Let μ be a (signed) Borel measure on Ω , which is finite on compact subsets of Ω (Radon measure). Then

$$T_{\mu}(\varphi) = \langle T_{\mu} | \varphi \rangle_{D^*, D} = \int_{\Omega} \varphi(x) \, d\mu(x)$$

is a distribution. A concrete example of it is if we take $\Omega = \mathbb{R}$, $\mu = \delta_0 = \#_{\{0\}}$, which gives us the **Dirac Delta** (Dirac Mass)

$$\delta_0(\varphi) = \langle \delta_0 | \varphi \rangle_{D^*, D} = \int_{\mathbb{R}} \varphi(x) \, d\delta_0(x) = \varphi(0)$$

Observe that if $f \in C^\infty(\Omega)$, $\varphi \in \mathcal{D}(\Omega)$, we can always use integration by parts

$$\int (\partial_x^\alpha f) \varphi \, dx = (-1)^{|\alpha|} \int f (\partial_x^\alpha \varphi) \, dx,$$

which motivates the following definition.

1.64 Definition [Weak Derivative]

Let $f \in L^1_{\text{loc}}(\Omega)$, if there exists $g \in L^1_{\text{loc}}(\Omega)$, such that

$$\int f (\partial_x^\alpha \varphi) \, dx = (-1)^{|\alpha|} \int g \varphi \, dx, \quad \forall \varphi \in \mathcal{D}(\Omega).$$

Then, g is called a weak derivative of f of order α . Can also be defined for L^2 functions.

1.65 Definition [Distributional Derivative]

Let T be a distribution, then its **distributional derivative** of order α is defined via transpose

$$\langle \partial_x^\alpha T \mid \varphi \rangle := (-1)^{|\alpha|} \langle T \mid \partial_x^\alpha \varphi \rangle$$

1.66 Remark

Distributional, weak, classical derivatives are related as following. A weak derivative $\partial^\alpha f = g$ coincides with the classical derivative (if exists). The corresponding distributions T_f, T_g are thus related by the distributional derivative $\partial^\alpha T_f = T_g$

1.67 Example

Let $H : \mathbb{R} \rightarrow \mathbb{R}; x \mapsto \mathbf{1}_{\mathbb{R}_+}$. It has a distributional derivative Dirac δ_0 but has neither classical nor weak derivative.

1.68 Definition [Operations on Distributions]

Let $F \in \mathcal{D}^*, \varphi \in \mathcal{D}, \psi \in C^\infty$. Let $\tau_h f(x) = f(x - h)$ be a translation operator, let $f_a(x) = f(ax)$ be dilation and $f_L(x) = f(L(x))$ be a non-singular linear operator. Then, we define operators on S^* via transposes,

$$\begin{aligned} \langle \psi \cdot F \mid \varphi \rangle &:= \langle F \mid \psi \cdot \varphi \rangle \\ \langle \psi \star F \mid x \mapsto \varphi(-x) \rangle &:= \langle F \mid \psi \star \varphi \rangle \\ \langle \tau_h F \mid \varphi \rangle &:= \langle F \mid \tau_{-h} \varphi \rangle \\ \langle F_a \mid \varphi \rangle &:= a^{-d} \langle F \mid \varphi_{a^{-1}} \rangle \\ \langle F_L \mid \varphi \rangle &:= |\det L|^{-1} \langle F \mid \varphi_{L^{-1}} \rangle \end{aligned}$$

1.69 Definition [Convolution]

Let $\varphi \in \mathcal{D}(\mathbb{R}^d)$ and $T, S \in \mathcal{D}^*(\mathbb{R}^d)$

1.70 Definition [Convergence of Test Functions]

$(\varphi_n) \subset \mathcal{D}$ weakly converges to some $\varphi \in \mathcal{D}$

$$\varphi_n \xrightarrow[n \rightarrow \infty]{\sigma(\mathcal{D}, \mathcal{D}^*)} \varphi \qquad \varphi_n \xrightarrow{\mathcal{D}} \varphi$$

iff

$$\langle \Phi | \varphi_n \rangle \xrightarrow[n \rightarrow \infty]{} \langle \Phi | \varphi \rangle, \quad (\forall \Phi \in \mathcal{D}^*)$$

This is a very strong notion of convergence and we rarely care about the sequences of test functions and its strong and weak limits, so it is **never used**.

1.71 Definition [Convergence of Distributions]

$(T_n) \subset \mathcal{D}^*$ weakly^{*} converges to some $T \in \mathcal{D}^*$

$$T_n \xrightarrow[n \rightarrow \infty]{\sigma(\mathcal{D}^*, \mathcal{D})} T \quad T_n \xrightarrow{\mathcal{D}^*} T$$

iff

$$\langle T_n | \varphi \rangle \xrightarrow[n \rightarrow \infty]{} \langle T | \varphi \rangle, \quad (\forall \varphi \in \mathcal{D})$$

This is the weakest notion of convergence and is used very frequently.

Convergence in the sense of distributions is what we call when we take a sequence, e.g. $(u_n) \subset L^2$, and we look at the copy of this sequence in $\mathcal{D}^*$. As each u_n defines a distribution $T_{u_n} \in \mathcal{D}^*$, we can look at the weak^{*} limit of this sequence

$$T_{u_n} \xrightarrow{\mathcal{D}^*} T,$$

and then we can say (u_n) converges to $T \in \mathcal{D}^*$, in the sense of distributions. A posteriori, the limiting distribution T might also contain a copy in a more structured space, e.g. $u \in L^2$.

1.72 Proposition

We have the following chain of inclusions

$$D \subsetneq L^p \subsetneq L_{\text{loc}}^p \subsetneq L_{\text{loc}}^1 \subsetneq \mathcal{D}^*$$

1.10 Sobolev Spaces of Integer Power

1.73 Definition [Sobolev Spaces $W^{m,p}$]

Let $\Omega \subset \mathbb{R}^d$ be open. Let $m \in \mathbb{N}, p \in [1, \infty]$. Then, we define $W^{m,p}(\Omega)$ to be the of of

$L^p(\Omega)$ functions, whose distributional derivatives up to order m are in L^p

$$W^{m,p}(\Omega) = \{u \in L^p; \forall |\alpha| \leq m, \partial^\alpha u \in L^p\}$$

With the norm

$$\|u\|_{W^{m,p}} = \sum_{|\alpha| \leq m} \|\partial^\alpha u\|_{L^p}$$

and seminorm

$$[u]_{W^{m,p}} = \sup_{|\alpha|=m} \|\partial^\alpha u\|_{L^p}$$

Functional Inequalities, Embeddings

1.74 Theorem [Hölder Inequality]

Let $p, q, r \in [1, \infty]$, $f \in L^p(\Omega)$, $g \in L^q(\Omega)$ and

$$\frac{1}{p} + \frac{1}{q} = \frac{1}{r}$$

Then,

$$\|fg\|_{L^r} \leq \|f\|_{L^p} \|g\|_{L^q},$$

which implies that $fg \in L^r(\Omega)$.

(p, q) is called Hölder pair, p and q are each other's Hölder conjugates.

1.75 Theorem [Interpolation Inequality]

Let $p_0 < p < p_1$ with

$$\frac{1}{p} = \frac{\theta}{p_0} + \frac{1-\theta}{p_1}$$

Then,

$$\|f\|_p \leq \|f\|_{p_0}^\theta \|f\|_{p_1}^{1-\theta}$$

1.76 Proposition [Density of Test Functions]

Let Ω be Lipschitz, then, if we take the set of test functions, restricted to Ω and then

take its closure w.r.t. the norm of the respective Sobolev space, we will get

$$\overline{\mathcal{D}_{|\Omega}(\mathbb{R}^d)}^{\|\cdot\|_{W^{m,p}}} = W^{m,p}(\Omega).$$

1.77 Theorem [Poincaré Inequality]

Let Ω be Lipschitz and bounded in at least one direction, $p \in [1, \infty]$, $u \in W_0^{1,p}$. Then,

$$\|u\|_{L^p} \lesssim \|\nabla u\|_{L^p}$$

One consequence of that is it makes $\|\nabla u\|_{L^p}$ equivalent to the Sobolev norm on spaces, on which the Poincaré inequality holds.

1.78 Theorem [Gagliardo–Nirenberg–Sobolev Embeddings]

Let $\Omega \subset \mathbb{R}^d$ and let m, d, p, p^* satisfy

$$\frac{1}{p^*} := \frac{1}{p} - \frac{m}{d} > 0.$$

Then, we have a continuous embedding

$$W^{m,p}(\Omega) \hookrightarrow L^{p^*}$$

with

$$\|u\|_{L^{p^*}(\mathbb{R}^d)} \lesssim [u]_{W^{m,p}(\mathbb{R}^d)}$$

and generally

$$\|u\|_{L^{p^*}(\Omega)} \lesssim [u]_{W^{m,p}(\Omega)} + \|u\|_{L^p(\Omega)}$$

1.79 Theorem [Morrey]

Let $u \in W^{1,p}(\mathbb{R}^d)$ with $p > d$, then

$$W^{1,p}(\mathbb{R}^d) \hookrightarrow C^{0,\gamma}(\mathbb{R}^d), \quad \|u\|_{C^{0,\gamma}(\mathbb{R}^d)} \lesssim \|u\|_{W^{1,p}(\mathbb{R}^d)}.$$

with $\gamma = 1 - \frac{d}{p}$

1.80 Theorem [Rellich–Kondrashov]

Let Ω be bounded, Lipschitz, let

$$\frac{1}{p} - \frac{m}{d} < \frac{1}{q} \leq \frac{1}{p}$$

Then,

$$W^{m,p}(\Omega) \Subset L^q(\Omega)$$

In particular, $W^{m,p} \Subset L^p$

1.81 Theorem [Gagliardo–Nirenberg]

Let $j, m \in \mathbb{N}$ and $p, q, r \in [1, \infty]$ with $j < m$. If

$$\exists \theta \in [j/m, 1) : \frac{1}{p} - \frac{j}{d} = \theta \left(\frac{1}{r} - \frac{m}{d} \right) + \frac{1-\theta}{q}$$

Then,

$$[u]_{W^{j,p}(\mathbb{R}^d)} \lesssim [u]_{W^{m,r}(\mathbb{R}^d)}^\theta \|u\|_{L^q(\mathbb{R}^d)}^{1-\theta}$$

1.82 Proposition [Product Rule]

Let $u, v \in W^{m,p}(\mathbb{R}^d)$ with $mp > d$. Then,

$$\|uv\|_{L^\infty} \lesssim \|u\|_{W^{1,\infty}} \|v\|_{W^{m,p}} + \|u\|_{W^{m,p}} \|v\|_{W^{1,\infty}}$$

Compactness and Defects of Compactness

1.83 Theorem [Weak Compactness in Banach Spaces]

Let X be a reflexive Banach space. Let $(x_n)_n \subset X$ be bounded. Then, there exists a subsequence $(x_{n_k})_k$ that weakly converges to some $x \in X$

$$x_{n_k} \xrightarrow{X} x$$

$$\langle \ell | x_{n_k} \rangle_{X^*, X} \xrightarrow[k \rightarrow \infty]{} \langle \ell | x \rangle_{X^*, X}, \quad \forall \ell \in X^*$$

Moreover,

$$\|x\|_X \leq \liminf_k \|x_{n_k}\|_X$$

meaning that, in the weak limit, due to the defects of compactness, mass can escape.

1.84 Theorem [Weak* Compactness in Banach Spaces]

Let X be a separable Banach space. Let $(\ell_n)_n \subset X^*$ be bounded. Then, there exists a subsequence $(\ell_{n_k})_k$ that weakly* converges to some $\ell \in X^*$

$$\ell_{n_k} \xrightarrow{X^*} \ell$$

$$\langle \ell_{n_k} | x \rangle_{X^*, X} \xrightarrow{k \rightarrow \infty} \langle \ell | x \rangle_{X^*, X}, \quad \forall x \in X$$

Moreover,

$$\|\ell\|_{X^*} \leq \liminf_k \|\ell_{n_k}\|_{X^*}$$

meaning that, in the weak* limit, due to the defects of compactness, mass can escape.

1.85 Proposition

Let $(u_n)_n \subset W^{m,p}$ be bounded. Then, there exists an extraction $(u_{n_k})_n$ and some $u \in W^{m,p}$, such that $\forall |\alpha| \leq m$:

$$\partial^\alpha u_{n_k} \xrightarrow{L^2} \partial^\alpha u, \quad \|u\|_{W^{m,p}} \leq \liminf_{k \rightarrow \infty} \|u_{n_k}\|_{W^{m,p}}$$

1.86 Proposition

Let H be Hilbert, $(u_n)_n \subset H$. If $u_n \xrightarrow{H} u$, then,

$$\|u_n - u\|_H \xrightarrow{n \rightarrow \infty} 0 \iff \|u_n\|_H \xrightarrow{n \rightarrow \infty} \|u\|_H$$

1.87 Proposition [Brézis–Lieb Lemma]

Let $p \in [1, \infty)$ and $(u_n)_n \subset L^p(\Omega)$. If $u_n \xrightarrow[n \rightarrow \infty]{a.e.} u$, then,

$$\|u_n - u\|_{L^p} \xrightarrow{n \rightarrow \infty} 0 \iff \|u_n\|_{L^p} \xrightarrow{n \rightarrow \infty} \|u\|_{L^p}$$

Moreover, a stronger result holds

$$\int_{\Omega} \left| |u_n(x)|^p - |u(x)|^p - |u_n(x) - u(x)|^p \right| dx \xrightarrow{n \rightarrow \infty} 0$$

1.88 Proposition

Let $(u_n)_n \subset H^1(\Omega)$ be a bounded sequence. Then, (by Banach–Alaoglu) there exists an extraction $(u_{n_k})_k$ and some $u \in H^1$ such that

$$u_{n_k} \xrightarrow{H^1} u$$

If, additionally Ω bounded and Lipschitz, then (by Rellich–Kondrashev), $(u_n)_n$ is pre-compact in $L^2(\Omega)$

$$\|u_{n_k} - u\|_{L^2(\Omega)} \xrightarrow{k \rightarrow \infty} 0$$

1.89 Theorem [Lions Concentration-Compactness principle]

1.90 Proposition

Let $(u_n) \xrightarrow{L^p} u$, $(v_n) \xrightarrow{(L^p)^*} v$. Then we still can have a defect of compactness so that

$$u_n v_n \not\xrightarrow{\mathcal{D}^*} uv$$

This can happen due to the defects of compactness in the forms of: oscillations, vanishing, concentration, and splitting.

1.91 Theorem

Let $p \in [1, \infty]$, $(u_n) \xrightarrow[n \rightarrow \infty]{L^p} u$ and $v_n \xrightarrow{(L^p)^*} v$. Then,

$$\int u_n v_n \xrightarrow{n \rightarrow \infty} \int uv$$

Let $(u_n) \xrightarrow{L^p} u, (v_n) \xrightarrow{(L^p)^\star} v$ with (u_n) bounded in $W^{1,p}(\mathbb{R}^d)$, (v_n) bounded in $(L^p)^\star$. Then, for any φ bounded and compactly supported,

$$\int_{\mathbb{R}^d} u_n v_n \varphi \xrightarrow{n \rightarrow \infty} \int_{\mathbb{R}^d} u v \varphi$$

1.92 Theorem [Div-Curl Lemma]

Let $(u_n), (v_n) \subset (L^p(\mathbb{R}^d))^d$ be bounded sequences of vector fields with

- 1) $u_n \xrightarrow{L^p} u, v_n \xrightarrow{L^p} v$
- 2) $(\operatorname{div} u_n), (\operatorname{curl} v_n)$ are L^p bounded.

Then,

$$u_n v_n \xrightarrow{\mathcal{D}^\star} uv$$

The proof uses the Helmholtz decomposition in the Fourier space.

1.93 Proposition [Weak Continuity of the Determinant]

Let $(u_n) \subset (W^{1,d}(\mathbb{R}^d))^d$ be a bounded sequence of vector fields. If

$$u_n \xrightarrow{W^{1,d}} u$$

Then,

$$\det(\nabla u_n) \xrightarrow{\mathcal{D}^\star} \det(\nabla u)$$

1.94 Theorem [Compactness with a chain of spaces]

Let $E_1 \subset E_2 \subset E_3$ Banach spaces with E_1 compactly embedded into E_2 , let $p \in [1, \infty]$. If a bounded subset $A \subset L^p([0, T], E_1)$ has the equicontinuity property

$$\sup_{f \in A} \|f(x+h) - f(x)\|_{L^p_x([0, T-h], E_3)} \xrightarrow{h \rightarrow 0} 0$$

then, A is relatively compact in $L^p([0, T], E_2)$

1.95 Proposition [Aubin–Lions Lemma]

Let X Banach, let $u \in L^p([0, T]; X)$, define for $a > 0$ the right mean function $M_a \in C([0, T - a]; X)$

$$M_a(f)(t) = \frac{1}{a} \int_t^{t+a} f(s) \, ds$$

Then, for $p \leq q$

$$\|M_a(f)\|_{L^q([0, T-a])} \leq a^{1/q-1/p} \|f\|_{L^p([0, T])}$$

2.0 Harmonic Analysis

2.1 Schwartz Space

2.1 Definition [Schwartz Space $\mathcal{S}$]

Let $n \in \mathbb{N}, \alpha \in \mathbb{N}^d$. The Schwartz space is defined as the set

$$\mathcal{S}(\mathbb{R}^d) := \{\varphi \in C^\infty ; (\forall \alpha : \forall k) \quad [\varphi]_{\alpha,k} < \infty\}$$

where the seminorms are defined as

$$[\varphi]_{\alpha,k} := \sup_{x \in \mathbb{R}^d} (1 + |x|^2)^k |\partial^\alpha \varphi(x)|$$

This space is a Fréchet space with the Schwartz topology, however one rarely needs strong convergence results.

2.2 Definition [Tempered Distributions $\mathcal{S}^*$]

The space of tempered distributions is defined to be a topological dual of a Schwartz space

$$\mathcal{S}^* := (\mathcal{S})^*$$

It is the set of distributions that are continuous in the following sense

$$(\forall T \in \mathcal{S}^* : \exists n, A, C : \forall \varphi \in \mathcal{S}) \quad |T(\varphi)| \leq C \sup_{|\alpha| \leq A} [\varphi]_{\alpha,n}$$

We have some notions of strong topologies on both $\mathcal{S}, \mathcal{S}^*$ but in practice the derived $\sigma(\mathcal{S}, \mathcal{S}^*), \sigma(\mathcal{S}^*, \mathcal{S})$ topologies will be used.

2.3 Proposition

We have the following chain of inclusions over $p \in [1, \infty]$ on $\mathbb{R}^d$:

$$\mathcal{D} \subsetneq \mathcal{S} \subsetneq L^p \subsetneq \mathcal{S}^* \subsetneq \mathcal{D}^*$$

With non-examples:

- 1) $e^{-|x|^2} \in \mathcal{S}(\mathbb{R}^d)$, but not in $\mathcal{D}(\mathbb{R}^d)$

- 2) $(x \mapsto 1) \in L^\infty$, but not in $\mathcal{S}$
- 3) $\delta_0 \in \mathcal{S}^*$, but not in L^p
- 4) $e^x \in \mathcal{D}^*(\mathbb{R})$, but not in $\mathcal{S}^*(\mathbb{R})$

2.2 Fourier Transform

2.4 Definition [Fourier Transform]

Let $f \in \mathcal{S}(\mathbb{R}^d; \mathbb{C})$, the Fourier transform is a linear mapping

$$\begin{aligned} \mathcal{F} : \mathcal{S}(\mathbb{R}^d; \mathbb{C}) &\rightarrow \mathcal{S}(\mathbb{R}^d; \mathbb{C}) \\ f &\mapsto \widehat{f}(\xi) = \int_{\mathbb{R}^d} e^{-i x \cdot \xi} f(x) \, dx \end{aligned}$$

2.5 Theorem

The Fourier transform is invertible with the inverse

$$\mathcal{F}^{-1} = (2\pi)^{-d} \mathcal{F}^*$$

with

$$\mathcal{F}^*[f](\xi) = \int_{\mathbb{R}^d} e^{+i x \cdot \xi} f(x) \, dx$$

The Fourier transform is an isometric isomorphism on $\mathcal{S}(\mathbb{R}^d; \mathbb{C})$.

2.6 Theorem

Fourier transform is self-transposed and we can be thus defined as generally as on the space of tempered distributions

$$\mathcal{F} : \begin{array}{ccc} \mathcal{S}^*(\mathbb{R}^d; \mathbb{C}) & \longrightarrow & \mathcal{S}^*(\mathbb{R}^d; \mathbb{C}) \\ T & \longmapsto & \mathcal{F}T \end{array}$$

via transposition

$$\langle \mathcal{F}T \mid f \rangle_{\mathcal{S}^*, \mathcal{S}} = \langle T \mid \mathcal{F}f \rangle_{\mathcal{S}^*, \mathcal{S}} \quad \forall f \in \mathcal{S}$$

2.7 Theorem [Fourier-Plancherel, Parseval]

Let $f \in \mathcal{S}(\mathbb{R}^d)$. Then,

$$(2\pi)^d (f | g)_{L^2} = (\widehat{f} | \widehat{g})_{L^2}$$

and, in particular,

$$(2\pi)^d \|f\|_{L^2} = \|\widehat{f}\|_{L^2}$$

2.8 Theorem [Extension to L^2]

By the extension of linear maps, $\mathcal{F}$ can be extended to $L^2(\mathbb{R}^d; \mathbb{C})$, and by Fourier-Plancherel, $\mathcal{F}$ is an isometric isomorphism on L^2 . Note that if $(f \in L^2, f \notin L^1)$, there will be no explicit formula to compute $\widehat{f}$.

By density, $\mathcal{F}$ extends to L^p spaces for $p \in [1, \infty)$.

2.9 Remark

Fourier transform and its combinations appear in many fields and problems

Domain	Space	→	Fourier Domain (Space)
(x)	Position space	→	(ξ) : Spatial frequency space
(t)	Time domain	→	(τ) : Frequency space
(t, x)		→	(τ, ξ) : Space-time frequency space
(t, x)	Physical space	→	(t, ξ) : Momentum space (spatial Fourier space)
(t, x)		→	(τ, x) : Spectral space (temporal Fourier space)

2.3 Fourier Operators

2.10 Definition [Fourier Multiplier]

An operator T_m is a **Fourier (Multiplier) Operator** if there is a function $m(\xi)$ called a **Fourier Symbol**, such that it has a following form

$$T_m := m(\nabla/i) := \mathcal{F}^{-1} m(\xi) \mathcal{F}$$

$$\mathcal{F}[T_m f(x)] = \mathcal{F}[m(\nabla/i) f(x)] = m(\xi) \mathcal{F} f = m(\xi) \widehat{f}(\xi)$$

$\mathcal{F}^{-1} m(\xi) \mathcal{F} f$ should be understood as $\mathcal{F}^{-1} (m(\xi) \mathcal{F}[f])$.

2.11 Example [Fourier Operators]

Let $k \in \mathbb{N}^\times, s \in \mathbb{C}$

Derivative $\partial_x^k = \mathcal{F}^{-1}(i\xi)^k \mathcal{F}$

$$\mathcal{F}[\partial_x^k f(x)] = \mathcal{F}[\mathcal{F}^{-1}(i\xi)^k \mathcal{F}f(x)] = (i\xi)^k \widehat{f}(\xi)$$

Fractional (Riesz) derivative $D_x^s := |\partial_x|^s := \mathcal{F}^{-1}|\xi|^s \mathcal{F}$

$$\mathcal{F}[D_x^s f(x)] = \mathcal{F}[\mathcal{F}^{-1}|\xi|^s \mathcal{F}f(x)] = |\xi|^s \widehat{f}(\xi)$$

Hilbert transform $\mathcal{H} = \mathcal{F}^{-1}(-i \operatorname{sgn}(\xi)) \mathcal{F}$

$$\mathcal{F}[\mathcal{H}f(x)] = \mathcal{F}[\mathcal{F}^{-1}(-i \operatorname{sgn}(\xi)) \mathcal{F}f(x)] = -i \operatorname{sgn}(\xi) \widehat{f}(\xi)$$

The connection between ordinary derivative and Riesz derivative is

$$\partial_x^k = (D_x \mathcal{H})^k = (\mathcal{H} D_x)^k.$$

Riesz derivative is sometimes written in the form of a fractional Laplacian

$$|\partial_x|^2 = (-\Delta)^{s/2}.$$

Japanese bracket $\langle D_x \rangle^s = \mathcal{F}^{-1}(1 + |\xi|^2)^{s/2} \mathcal{F}$

$$\mathcal{F}[\langle D_x \rangle^s f(x)] = \mathcal{F}[\mathcal{F}^{-1}\langle \xi \rangle^s \mathcal{F}f(x)] = \langle \xi \rangle^s \widehat{f}(\xi)$$

2.12 Definition [Smooth Cut-off Function]

A function $\eta(x) \in C^\infty(\mathbb{R})$ is called a **smooth cut-off** function (or a **bump** function) supported on $[-1, 1]$ iff

$$\eta(x) \in [0, 1]; \quad \eta(x) = 1, \quad x \in \left[-\frac{1}{2}, \frac{1}{2}\right]; \quad \eta(x) = 0, \quad x \in \mathbb{R} \setminus [-1, 1]$$

2.4 Sobolev–Fourier Spaces

2.13 Definition [Sobolev–Fourier Spaces]

Let $s \in \mathbb{R}$, we define

$$H^s(\mathbb{R}^d) = \{u \in \mathcal{S}'(\mathbb{R}^d); \|u\|_{H^s} < \infty\}$$

with

$$\|u\|_{H^s} := \|\langle \xi \rangle^s \widehat{u}(\xi)\|_{L^2} = \left(\int_{\mathbb{R}^d} |\widehat{u}(\xi)|^2 (1 + |\xi|^2)^s d\xi \right)^{1/2}$$

2.14 Proposition

Fourier transform is an isometric isomorphism between Hilbert spaces

$$H^s(\mathbb{R}^d) \stackrel{\mathcal{F}}{\cong} L^2(\mathbb{R}^d, (1 + |\xi|^2)^s d\xi)$$

2.15 Proposition

Let $s \in \mathbb{R}$. Then, $\mathcal{S}(\mathbb{R}^d)$ is dense in $H^s(\mathbb{R}^d)$

2.16 Proposition [Duality]

Let $s \in \mathbb{R}$. Then, $(H^s(\mathbb{R}^d))^ = H^{-s}(\mathbb{R}^d)$*

2.17 Theorem [Embeddings]

Let $s > d/2$. Then, H^s is continuously embedded into the space of functions that tend to 0 at infinities.

Let $0 \leq s < d/2$. Then, H^s is continuously embedded into $L^{\frac{2d}{d-2s}}$ and

$$\|f\|_{L^p} \lesssim \|f\|_{\dot{H}^s}$$

Trace, Extension

2.18 Theorem [Trace, Lifting]

Let $\Omega \subset \mathbb{R}^d$ be a domain with Lipschitz boundary, let $u \in H^1(\Omega)$. Then, there exists a **Trace operator** in $\mathcal{L}(H^1(\Omega), H^{\frac{1}{2}}(\partial\Omega))$

$$\begin{aligned} \gamma_0 : \quad H^1(\Omega) &\longrightarrow H^{\frac{1}{2}}(\partial\Omega) \\ u &\longmapsto \gamma_0 u = "u|_{\partial\Omega}" \end{aligned}$$

$\partial\Omega$ is a null set w.r.t. Lebesgue measure on $\mathbb{R}^d$, thus it's impossible to directly restrict an H^1 function to it. Instead, we consider $\varphi \in C^\infty(\overline{\Omega})$ and obtain an estimate for continuity of $\gamma_0 : C^\infty(\overline{\Omega}) \rightarrow H^{\frac{1}{2}}(\partial\Omega)$

$$\|\gamma_0(\varphi)\|_{H^{\frac{1}{2}}(\partial\Omega)} \lesssim \|\varphi\|_{H^1(\Omega)}, \quad \forall \varphi \in (C^\infty(\overline{\Omega}), \|\cdot\|_{H^1(\Omega)})$$

C^∞ is dense in H^1 and thus we can extend trace to the whole space H^1 .

Thus, we lose $1/2$ of derivative by reducing the dimension.

The results extends to $H^s(\mathbb{R}^d) \rightarrow H^{s-\frac{1}{2}}(\mathbb{R}^{d-1})$ for $s > 1/2$, and can be obtained by extension from the Schwartz class.

Moreover, trace admits an explicit right inverse L ,

$$\gamma_0 \circ L = \mathbf{1}_{H^{s-\frac{1}{2}}(\mathbb{R}^{d-1})}$$

2.19 Theorem [Extension by 0]

Let $u \in H^s(\mathbb{R}_+^*)$. The extension by 0 as a mapping $H^s(\mathbb{R}_+^*) \rightarrow H^s(\mathbb{R})$ is continuous iff $s < 1/2$. For $s = 1/2$ iff

$$\int_0^\infty \frac{|u(x)|^2}{x} dx < \infty$$

We denote by $H_{00}^{\frac{1}{2}}$ the space of functions that can be extended by 0, it's a complete space for the norm

$$\|u\|_{H_{00}^{1/2}} = \int_0^\infty \frac{|u(x)|^2}{x} dx + \|u\|_{H^{1/2}}$$

2.20 Definition [Zero-trace Sobolev Spaces]

Let $s > 0$, we define

$$H_0^s(\Omega) = \overline{\mathcal{D}(\Omega)}^{H^s(\Omega)}$$

2.21 Proposition

Let $s > 0$. Then,

$$(H^s(\Omega))^* = H_0^{-s}(\Omega)$$

Moreover, if $0 < s < 1/2$, then

$$H_0^s(\Omega) = H^s(\Omega)$$

if $s = 1/2$, then γ_0 is continuous but not surjective

$$H_0^{1/2}(\Omega) \subsetneq H^{1/2}(\Omega)$$

if $s > 1/2$, then the trace operator γ_0 is well defined and continuous and so we have the usual characterization

$$H_0^s(\Omega) = \{u \in H^s(\Omega) ; \gamma_0(u) = 0\}$$

Results

2.22 Proposition [Stein Derivative]

Let $s \in (0, 1)$, then the $H^s(\mathbb{R}^d)$ norm has this equivalent

$$\|u\|_{H^s} \sim \|u\|_{L^2} + \left(\iint_{(\mathbb{R}^d)^2} \frac{|u(x+h) - u(x)|^2}{|h|^{d+2s}} dx dh \right)^{1/2}$$

2.23 Theorem [Rellich]

Let $s, R > 0$. Denote by $H_R^s(\mathbb{R}^d)$ the space of $H^s(\mathbb{R}^d)$ functions which vanish outside of $B_{\mathbb{R}^d}(0, R)$. Then, for any bounded subset $A \subset H_R^s(\mathbb{R}^d)$

$$\overline{A}^{L^2(\mathbb{R}^d)} \text{ is compact in } L^2(\mathbb{R}^d)$$

2.5 Interpolation

2.24 Definition [Distribution Function]

Let $f \in L^p(\mathbb{R}^d)$, $\lambda := \text{Leb}_{\mathbb{R}^d}$ the distribution function of f is

$$m_f(t) := \lambda(|f| < t)$$

It's always well-defined by Chebyshev inequality.

2.25 Definition [Decreasing Rearrangement]

Let f be such that

$$m_f(t) \xrightarrow[t \rightarrow \infty]{} 0$$

Then, we can define $f^* : \mathbb{R}_+^* \rightarrow \mathbb{R}_+$ as

$$f^*(r) = \int_0^\infty \mathbf{1}_{r \in [0, m_f(t))} dt = \inf_{s \in \mathbb{R}_+^*} \{m_f(s) \leq r\}$$

2.26 Definition [Lorentz Spaces]

Let $p \in [1, \infty)$, weak L^p spaces are defined to be

$$L^{p,\infty}(\mathbb{R}^d) := \{f ; \sup_{t \in \mathbb{R}_+^*} t^p m_f(t) < \infty\}$$

Its generalization, Lorentz Spaces are defined to be

$$L^{p,q}(\mathbb{R}^d) := \{f ; \left\| f^*(t) t^{1/p} \right\|_{L^q(\mathbb{R}^d, dt/t)} < \infty\}$$

2.27 Definition [Sublinear Operator]

An operator T is called sublinear iff

$$|T(cf)| = |c| |T(f)|, \quad |T(f+g)| \leq |T(f)| + |T(g)|$$

2.28 Definition [Strong Type]

Let $p, q \in [1, \infty]$, a sublinear operator $T : L^p \rightarrow L^q$ is called **of (strong) type (p, q)**

iff

$$\|Tf\|_{L^q} \lesssim \|f\|_{L^p}$$

2.29 Definition [Weak Type]

Let $p, q \in [1, \infty]$, a sublinear operator $T : L^p \rightarrow L^q$ is called **of weak type** (p, q) iff

$$\|Tf\|_{L^{q,\infty}} \lesssim \|f\|_{L^p}$$

For $q = \infty$, notions of weak and strong type coincide.

2.30 Theorem [Hölder Interpolation Inequality (log-convexity of $\|\cdot\|_{L^p}$)]

Let $p_0, p_1 \in [1, \infty]$ and $\theta \in [0, 1]$ with

$$\frac{1}{p_\theta} = \frac{\theta}{p_0} + \frac{1-\theta}{p_1}, \quad \text{for some } \theta \in [0, 1]$$

Then,

$$\|f\|_{L^{p_\theta}} \lesssim \|f\|_{L^{p_0}}^\theta \|f\|_{L^{p_1}}^{1-\theta}$$

2.31 Proposition

Let $f \in L^{p_\theta}$, then it can be written as a sum

$$f = g + h$$

with $g \in L^{p_0}, h \in L^{p_1}$

2.32 Theorem [Marcinkiewicz Interpolation]

Let T be a **sublinear** operator and $p_0, p_1, q_0, q_1 \in [1, \infty]$, $p_0 \neq p_1, q_0 \neq q_1$. If T is of **weak type** (p_0, q_0) and of **weak type** (p_1, q_1) , then, T is of **strong type** (p_θ, q_θ) with

$$\left(\frac{1}{p_\theta}, \frac{1}{q_\theta}\right) = \left(\frac{\theta}{p_0} + \frac{1-\theta}{p_1}, \frac{\theta}{q_0} + \frac{1-\theta}{q_1}\right), \quad \forall \theta \in (0, 1)$$

and moreover,

$$\|T\|_{L^{p_\theta} \rightarrow L^{q_\theta}} \lesssim M_0^\theta M_1^{1-\theta}$$

$$\|Tf\|_{L^{q_0,\infty}} \leq M_0 \|f\|_{L^{p_0}}, \quad \|Tf\|_{L^{q_1,\infty}} \leq M_1 \|f\|_{L^{p_1}}$$

2.33 Theorem [Weak Marcinkiewicz Interpolation]

2.34 Lemma [Hadamard Three-Lines Theorem]

let f holomorphic on $0 < \operatorname{Re}(z) < 1$, bounded and continuous on $0 < \operatorname{Re}(z) \leq 1$ with

$$M_0 = \sup_t |f(1+it)|, \quad M_1 = \sup_t |f(it)|$$

Then,

$$|f(\theta + it)| \leq M_0^\theta M_1^{1-\theta}, \quad \forall \theta \in [0, 1]$$

2.35 Theorem [Reisz–Thorin Complex Interpolation]

Let T be a linear operator and $p_0, p_1, q_0, q_1 \in [1, \infty]$. If T is simultaneously of strong type (p_0, q_0) and of (p_1, q_1) , then, T is of strong type (p_θ, q_θ) with

$$\left(\frac{1}{p_\theta}, \frac{1}{q_\theta}\right) = \left(\frac{\theta}{p_0} + \frac{1-\theta}{p_1}, \frac{\theta}{q_0} + \frac{1-\theta}{q_1}\right), \quad \forall \theta \in [0, 1]$$

and moreover,

$$\|T\|_{L^{p_\theta} \rightarrow L^{q_\theta}} \lesssim \|T\|_{L^{p_0} \rightarrow L^{q_0}}^\theta \|T\|_{L^{p_1} \rightarrow L^{q_1}}^{1-\theta}$$

2.36 Theorem [Stein Interpolation]

2.37 Theorem [Hausdorff–Young]

Let $p \in [1, 2]$, the Fourier transform extends continuously to a map $L^p(\mathbb{R}^d) \rightarrow L^{p'}(\mathbb{R}^d)$ for

$$\frac{1}{p} + \frac{1}{p'} = 1$$

with

$$\|\mathcal{F}\|_{L^p \rightarrow L^{p'}} \leq (2\pi)^{d/p'}$$

2.6 Hardy–Littlewood Maximal Function

2.38 Definition [Hardy–Littlewood Maximal Function]

Let $f \in L^1_{\text{loc}}(\mathbb{R}^d)$,

$$\mathcal{M}f(x) := \sup_{t>0} \frac{1}{|B(x, t)|} \int_{B(x, t)} |f(y)| \, dy$$

$\mathcal{M}$ is subadditive and it is continuous for $L^\infty \rightarrow L^\infty$ and $L^1 \rightarrow L^{1,\infty}$

2.39 Theorem [Hardy–Littlewood–Sobolev Inequality]

Let $\alpha \in (0, d)$ and $1 < p < r < \infty$ with

$$1 + \frac{1}{r} = \frac{1}{p} + \frac{\alpha}{d}$$

Then,

$$\| |\cdot|^\alpha \star f \|_{L^r(\mathbb{R}^d)} \lesssim \|f\|_{L^p(\mathbb{R}^d)}$$

2.7 Calderón–Zygmund Theory

2.40 Proposition [Calderón–Zygmund Decomposition]

Let $f \in L^1(\mathbb{R}^d)$ and $\lambda > 0$. Then, there exists a (finite or) countable family of cubes $(Q_j)_{j \in J}$ such that

$$\lambda < \frac{1}{|Q_j|} \int_{Q_j} f(x) dx \leq 2^d \lambda$$

and

$$\|f\|_{L^\infty(\mathbb{R}^d \setminus (\cup Q_j))} \leq \lambda$$

2.41 Definition [Calderón–Zygmund Convolution Kernel]

Calderón–Zygmund Convolution Kernel $K \in C^1(\mathbb{R}^d \setminus \{0\}; \mathbb{C})$ is a function, such that

$$|\nabla K| \lesssim |x|^{-(d+1)}$$

It also defines an associated (singular) Calderón–Zygmund operator $T : f \mapsto K \star f$ that is continuous $L^2 \rightarrow L^2$

2.42 Theorem [Calderón–Zygmund]

Calderón–Zygmund operators are continuous $L^p \rightarrow L^p$ for $p \in (1, \infty)$.

2.8 Multiplier Theory

2.43 Theorem [Mikhlin–Hörmander]

Let $m : \mathbb{R}^d \rightarrow \mathbb{C}$ be a symbol with

$$(\forall \alpha : |\alpha| \leq \frac{d}{2} + 1 : \exists C_\alpha : \forall \xi \in \mathbb{R}^d) : \quad |\xi|^{|\alpha|} |\partial^\alpha m(\xi)| \leq C_\alpha$$

Then, the Fourier multiplier operator

$$\begin{aligned} T_m : \mathcal{S} &\rightarrow \mathcal{S} \\ f &\mapsto \mathcal{F}^{-1} \left(m(\xi) \widehat{f} \right) = \mathcal{F}^{-1} m(\xi) \mathcal{F} f \end{aligned}$$

continuously extends to a bounded operator on $L^p(\mathbb{R}^d) \rightarrow L^p(\mathbb{R}^d)$ for $p \in (1, \infty)$

2.44 Proposition

The following Calderon–Zygmund operators are continuous $L^p \rightarrow L^p$ for $p \in (1, \infty)$

$$R_j : u \mapsto \mathcal{F}^{-1} \left(\frac{\xi_j}{|\xi|} \widehat{u} \right)$$

and

$$\partial_i \partial_j \Delta^{-1} : u \mapsto \mathcal{F}^{-1} \frac{\xi_i \xi_j}{|\xi|^2} \widehat{u}$$

In particular, it means that, for those spaces, Laplacian can control all the derivatives

up to 2

$$\|u\|_{W^{2,p}} \sim \|u\|_{L^p} + \|\Delta u\|_{L^p}$$

2.45 Definition [Bessel Potential Spaces]

Let $s \geq 0$, $p \in (1, \infty)$,

$$H^{s,p}(\mathbb{R}^d) := \{u \in L^p ; \langle (-\Delta)^{s/2} \rangle u = \mathcal{F}^{-1}(\langle \xi \rangle^s \hat{u}) \in L^p\}$$

2.9 Besov Spaces

2.46 Definition [Dyadic Partition of Unity]

2.47 Definition [Besov Spaces]

Besov Spaces are defined as

$$B_{p,q}^s(\mathbb{R}^d) \{u \in \mathcal{S}^*(\mathbb{R}^d) ; \|u\|_{B_{p,q}^s} < \infty\}$$

with the norm

$$\|u\|_{B_{p,q}^s} := \|2^{sj} \Delta_j u\|_{l_j^q L_x^p} = \left(\sum_{j \in \mathbb{N}} (2^{sj} \|\Delta_j u\|_{L_x^p})^q \right)^{1/q}$$

Estimates

2.48 Proposition

Let $p \in [1, \infty]$ and $j \in \mathbb{N}$. Then,

$$\|\Delta_j u\|_{L^p} \lesssim \|u\|_{L^p}$$

2.49 Theorem [Bernstein Inequalities]

Let $\lambda > 0$ and $1 \leq p \leq q \leq \infty$ and $k \in \mathbb{N}$. If $\text{supp}(\widehat{u}) \subset B(0, \lambda)$, then

$$\sup_{|\alpha|=k} \|\partial^\alpha u\|_{L^q} \lesssim \lambda^{k+d(1/p-1/q)} \|u\|_{L^p}$$

If $\text{supp}(\widehat{u}) \subset \{\lambda/2 \leq |\xi| \leq 2\lambda\}$, then

$$\sup_{|\alpha|=k} \|\partial^\alpha u\|_{L^p} \sim \lambda^k \|u\|_{L^p}$$

2.50 Theorem

$B_{p,q}^s$ is Banach for all exponents. $\mathcal{S} \subset B_{p,q}^s$. $\mathcal{D}$ is dense in $B_{p,q}^s$ for $p, q < \infty$

2.51 Proposition [Besov Embeddings]

Let $s \in \mathbb{R}$; $1 \leq p_1 \leq p_2 \leq \infty$; $1 \leq q_1 \leq q_2 \leq \infty$. Then,

$$\|u\|_{B_{p_2,q_2}^{s-d(1/p_1-1/p_2)}} \lesssim \|u\|_{B_{p_1,q_1}^s}$$

2.52 Proposition [Besov Interpolation]

Let $s_1 < s_2$; $1 \leq p, q \leq \infty$ and $\theta \in [0, 1]$. Then,

$$\|u\|_{B_{p,q}^{\theta s_1 + (1-\theta)s_2}} \leq \|u\|_{B_{p,q}^{s_1}}^\theta \|u\|_{B_{p,q}^{s_2}}^{1-\theta}$$

Paraproducts

2.53 Definition [Paraproduct]

Let u be fixed, the map

$$v \mapsto T_v u := \sum_{k \geq 2} S_{k-2} u \Delta_k v$$

Is called paraproduct of v by u , and the decomposition

$$uv = T_v u + T_u v + R(u, v)$$

is called the Bony decomposition

2.54 Theorem

Let $u \in L^\infty, v \in B_{p,q}^s$, $p, q \in [1, \infty], s \in \mathbb{R}$. Then,

$$\|T_u v\|_{B_{p,q}^s} \lesssim \|u\|_{L^\infty} \|v\|_{B_{p,q}^s}$$

and if

$$s_1 + s_2 > 0, \quad 1/p = 1/p_1 + 1/p_2, \quad 1/q = 1/q_1 + 1/q_2$$

we have

$$\|R(u, v)\|_{B_{p,q}^{s_1+s_2}} \lesssim \|u\|_{B_{p_1,q_1}^{s_1}} \|v\|_{B_{p_2,q_2}^{s_2}}$$

2.55 Proposition [Product Rule for Besov Spaces]

Let $u, v \in B_{p,q}^s \cap L^\infty$, $s > 0$. Then,

$$\|uv\|_{B_{p,q}^s} \lesssim \|u\|_{L^\infty} \|v\|_{B_{p,q}^s} + \|v\|_{L^\infty} \|u\|_{B_{p,q}^s}$$

2.10 Littlewood–Payley Theory

2.11 Oscillatory Integrals

2.56 Definition [Oscillatory Integrals]

Oscillatory integral of the first kind is defined to be an integral of the form

$$I(\lambda) = \int e^{i\lambda\phi(x)} \psi(x) \, dx$$

for $\lambda > 0, \phi : \mathbb{R}^d \rightarrow \mathbb{R}, \psi : \mathbb{R}^d \rightarrow \mathbb{C}$

Oscillatory integral of the second kind is defined as

$$T_\lambda f(x) = \int e^{i\lambda\phi(x,y)} K(x, y) f(y) \, dy$$

for $\lambda > 0, \phi : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}, K : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{C}$ and $\psi : \mathbb{R}^d \rightarrow \mathbb{C}$.

In both cases, we are interested in the asymptotic regime, as $\lambda \rightarrow \infty$

2.57 Proposition

2.58 Proposition [Van der Korput Lemma]

2.59 Proposition [Stationary Phase]

2.60 Lemma [Morse Lemma]

3.0 PDE Tools

3.1 General Results

3.1 Lemma [Grönwall]

Let I be an interval, $u \in C(I; \mathbb{R}_+)$ and $\beta \in L^1_{loc}(I; \mathbb{R})$ be integrable. If there exists $C \geq 0$ such that, for all $t \in I$,

$$u(t) \leq C + \int_{t_0}^t \beta(s)u(s) \, ds,$$

or equivalently if u is differentiable and

$$u'(t) \leq \beta(t)u(t), \quad t \in I, \quad u(t_0) = C,$$

then, for all $t \in I$,

$$u(t) \leq C \exp\left(\int_{t_0}^t \beta(s) \, ds\right).$$

3.2 Theorem [Cauchy–Lipschitz–Picard]

Let E Banach, and $F : E \rightarrow E$ be K -Lipschitz. **Then**, given any initial data u_0 , there exists a unique solution $u \in C^1([0, \infty); E)$ of the initial value problem

$$\left\{ \begin{array}{l} \partial_t u(t) = Fu(t) \quad t \in [0, \infty) \\ u(0) = u_0 \end{array} \right.$$

3.3 Definition [Unbounded Operator]

A linear operator $A : D(A) \subset H \rightarrow H$ is said to be

- 1) unbounded, if it is not necessarily bounded
- 2) monotone, if

$$(Av \mid v)_H \geq 0, \quad \forall v \in H$$

- 3) maximal monotone, if

$$I_H + A \text{ surjective}$$

3.4 Theorem [Hille–Yosida, Homogeneous]

Let H be Hilbert, let $A : D(A) \rightarrow E$ be a maximal monotone linear operator. **Then**, given any initial data $u_0 \in D(A)$, there exists a unique solution

$$u \in C^1([0, \infty]; H) \cap C([0, \infty]; D(A))$$

of

$$\begin{cases} \partial_t u + Au = 0 & t \in [0, \infty] \\ u(0) = u_0 \end{cases}$$

And moreover,

$$|u(t)| \leq |u_0|, \quad |\partial_t u(t)| = |Au(t)| \leq |Au_0|$$

3.5 Definition [Linear Propagators]

3.6 Theorem [Hille–Yosida, Inhomogeneous]

3.7 Theorem [Hille–Yosida, Nonlinear, Global]

3.8 Theorem [Hille–Yosida, Nonlinear, Local]

3.9 Theorem [Cauchy–Lipschitz–Picard on a real interval]

Let E be a Banach space, $U \subset \mathbb{R} \times E$ open, and $f : U \rightarrow E$. Fix $(t_0, x_0) \in U$ and strictly positive reals $\alpha_0, R_0, K > 0$. For $\alpha > 0$ denote $I_\alpha = [t_0 - \alpha, t_0 + \alpha]$ and $B(x_0, R_0)$. Assume that on $\mathcal{A} := I_{\alpha_0} \times B(x_0, R_0)$ the map f is continuous and Lipschitz in the second variable:

$$\|f(t, x) - f(t, y)\|_E \leq K\|x - y\|_E \quad \text{for all } (t, x), (t, y) \in \mathcal{A}.$$

Set

$$M_0 := \sup_{(t,x) \in \mathcal{A}} \|f(t, x)\|_E \quad (\text{finite by continuity on } \mathcal{A}),$$

and choose

$$\alpha := \min\left(\alpha_0, \frac{R_0}{M_0}, \frac{1}{K}\right).$$

Then there exists a unique $\gamma \in C^1(I_\alpha; E)$ with

$$\gamma(t) = x_0 + \int_{t_0}^t f(s, \gamma(s)) \, ds, \quad t \in I_\alpha,$$

hence $\dot{\gamma}(t) = f(t, \gamma(t))$ for all $t \in I_\alpha$ and $\gamma(t) \in B(x_0, R_0)$ for $t \in I_\alpha$.

3.2 Fixed Point Theorems

3.10 Theorem [Banach–Picard Fixed Point]

Contractive functions on complete metric spaces uniquely fix a point.

Let (X, d) be a non-empty complete metric space, $K < 1$,

$f : X \rightarrow X$ be a continuous mapping with property

$$d(f(x), f(y)) \leq K d(x, y), \quad \forall x, y \in X.$$

Then, there exists a unique $x_\star$, such that $f(x_\star) = x_\star$

*Such f is called a **strict contraction** and $x_\star$ is called a **fixed-point**.*

3.11 Theorem [Kato Fixed Point]

Let V, X be two Banach spaces with $V \hookrightarrow X$ continuously, V separable, reflexive,

$f : V \rightarrow V$, such that, for some $R > 0, k \in (0, 1)$ it holds that

$$(\forall u, v \in \overline{B_V}(0, R)) : \quad \|f(u)\|_V \leq R, \quad \|f(u) - f(v)\|_X \leq k \|u - v\|_X$$

Then, f has a unique fixed point in $\overline{B_V}(0, R)$

3.12 Theorem [Brouwer Fixed Point]

Let E be a **finite-dimensional** vector space, $C \subset E$ a compact, convex subset. If $T : C \rightarrow C$ is continuous, then T has a fixed point (not necessarily unique).

3.13 Theorem [Schauder Fixed Point]

Let E be a **normed** vector space, $C \subset E$ a compact, convex subset. If $T : C \rightarrow C$ is continuous, then T has a fixed point (not necessarily unique).

3.14 Lemma [Leray–Schauder fixed point]

Let E be banach, $T(x, \lambda) : E \times [0, 1] \rightarrow E$ a continuous map with

- 1) T is a compact mapping
- 2) $T(x, 0) \equiv 0$
- 3) $\exists R > 0 : \forall \lambda \in [0, 1] : T(x, \lambda) = x \implies \|x\| < R$

Then, $x \mapsto T(x, 1)$ has a fixed point (not necessarily unique).

3.3 Variational Formulation

3.15 Theorem [Green's Identity]

Let Ω with a Lipschitz boundary, let $u, v \in H^1(\Omega)$, and let $\gamma_0 : H^1(\Omega) \rightarrow H^{1/2}(\partial\Omega)$ be the trace operator and $\gamma_1 : H^1(\Omega) \rightarrow H^{-1/2}(\partial\Omega)$ the weak normal derivative, behaving as (actually defined more technically)

$$\gamma_0(u) = u|_{\partial\Omega}, \quad \gamma_1(u) = \partial_n u|_{\partial\Omega}$$

Then,

$$\int_{\Omega} v \Delta u \, dx = \int_{\partial\Omega} \gamma_0(v) \gamma_1(u) \, d\sigma - \int_{\Omega} \nabla u \cdot \nabla v$$

Let Ω be bounded, $u \in H^1(\Omega)$ and $f \in H^{-1}(\Omega)$. Then, the variational formulation of

the Poisson equation with Homogeneous boundary conditions

$$\left| \begin{array}{l} -\Delta u(x) = f(x), \quad x \in \Omega, \\ \gamma_0(u)(\sigma) = 0, \quad \sigma \in \partial\Omega \end{array} \right.$$

is the following, $\forall v \in H^1(\Omega)$,

$$a(u, v) = \ell(v)$$

Where, ℓ is a continuous linear form

$$\ell(v) = \langle f | v \rangle_{H^{-1}(\Omega), H^1(\Omega)} = \int_{\Omega} f v$$

and $a : H^1 \times H^1 \rightarrow \mathbb{R}$ is a continuous, coercive bilinear form

$$a(u, v) = \int_{\Omega} \nabla u \cdot \nabla v$$

By Lax-Milgram, it has a unique solution in $H^1(\Omega)$, with the norm controlled by

$$\|u\|_{H^1(\Omega)} \leq \frac{1}{\alpha} \|f\|_{H^{-1}(\Omega)}$$

where α is the coercivity (ellipticity) constant of the bilinear form.

For the more general case of inhomogeneous Robin boundary conditions (Dirichlet + Neumann), let $\Omega \subset \mathbb{R}^d$, $u \in H^1(\Omega)$ and $f \in H^{-1}(\Omega)$, $g \in H^{-1}(\partial\Omega)$ and n the outward normal vector on $\partial\Omega$ with $\gamma_1(u) = \partial_n u$

$$\left| \begin{array}{l} -\Delta u(x) + u = f(x), \quad x \in \Omega, \\ \gamma_1(u)(\sigma) + \gamma_0(u)(\sigma) = g(\sigma), \quad \sigma \in \partial\Omega \end{array} \right.$$

via

$$\int_{\Omega} \Delta u v = - \int_{\Omega} \nabla u \cdot \nabla v + \int_{\partial\Omega} \gamma_1(u) \gamma_0(v)$$

Energy Formulation

Given the variational formulation, we can define the energy functional

$$J(u) = \frac{1}{2} a(u, u) - \ell(u)$$

In the case of symmetric bilinear form a , the solution of the variational formulation is characterized as the unique ground state of the energy functional

$$u = \arg \min_{v \in H^1(\Omega)} J(v) = \arg \min_{v \in H^1(\Omega)} \frac{1}{2} a(v, v) - \ell(v)$$

Homogenization

$$\left| \begin{array}{l} - \left[a \left(\frac{x}{\varepsilon} \right) u'_\varepsilon(x) \right]' = f(x), \quad x \in (0, 1), \\ u_\varepsilon(0) = u_\varepsilon(1) = 0 \end{array} \right.$$

As $\varepsilon \rightarrow 0$, it goes to an effective equation

$$\left| \begin{array}{l} - [a_\star u'_\star(x)]' = f(x), \quad x \in (0, 1), \\ u_\star(0) = u_\star(1) = 0 \end{array} \right.$$

with

$$a_\star = \left\langle \frac{1}{a} \right\rangle^{-1} = \left(\int_0^1 \frac{1}{a(x)} dx \right)^{-1}$$

3.16 Proposition

Let $b \in L^\infty(\mathbb{R})$ with period 1 and $\phi \in L^1(\mathbb{R})$. Then,

$$\lim_{\varepsilon \rightarrow 0} \int_{\mathbb{R}} b \left(\frac{x}{\varepsilon} \right) \phi(x) dx = \langle b \rangle \int_{\mathbb{R}} \phi(x) dx$$

Some Things to Remember

3.17 Proposition

Let $u(x) : \mathbb{R}^d \rightarrow \mathbb{R}$ be radial of the form $f(r)$, $r = |x|_{\ell^2}$.

The Laplacian Δ then takes the scalar form

$$\Delta u(x) = f''(r) + \frac{d-1}{r} f'(r)$$

And the gradient has the form

$$\nabla u(x) = f'(r) \frac{1}{r} x$$

3.4 Harmonic Functions

4.0 Differential Equations

4.1 Laplace Equation

Let $u : \Omega \rightarrow \mathbb{R}$

$$\Delta u(x) = 0 \quad (\text{Laplace})$$

Nongomogeneous Laplace also known as Poisson equation

$$-\Delta u(x) = f(x) \quad (\text{Poisson})$$

Dirichlet Problem

Let $\Omega \subset \mathbb{R}^d$ with Lipschitz boundary.

$$\left| \begin{array}{ll} -\Delta u(x) = f(x), & x \in \Omega, \\ u(x) = 0, & x \in \partial\Omega \end{array} \right. \quad (1)$$

Intuition:

Laplace equation models the steady-state flows in equilibrium. Poisson equation adds a source potential.

Symmetries:

Solution:

Conservation Laws:

A variant of a Laplace equation is the related Helmholtz equation

$$-\Delta u(x) = k^2 u(x) \quad (\text{Helmholtz})$$

4.2 Heat Equation

Let $u : I \times \Omega \rightarrow \mathbb{R}$

$$\partial_t u = \Delta u \quad (\text{Heat})$$

If $\Omega = \mathbb{R}^d$, we have an initial value (Cauchy) problem for the heat equation on the whole space. Solving such a problem requires an initial condition.

$$\left| \begin{array}{ll} \partial_t u(t, x) = \Delta u(t, x) & (t, x) \in I \times \mathbb{R}^d \\ u(0, x) = u_0(x) & x \in \mathbb{R}^d \end{array} \right. \quad (\text{Free Heat})$$

If Ω is a regular bounded region, we need to specify a boundary condition.

$$\left| \begin{array}{ll} \partial_t u(t, x) = \Delta u(t, x) & (t, x) \in I \times \Omega \\ u(0, x) = u_0(x) & x \in \Omega \\ u(t, x) = g(t, x) & x \in \partial\Omega \end{array} \right. \quad (2)$$

Intuition:

Symmetries:

Solution:

Conservation Laws:

Nonhomogeneous heat equation

$$\partial_t u - \Delta u = f(t, x) \quad (3)$$

4.3 Wave Equation

Let $u : I \times \Omega \rightarrow \mathbb{R}$

$$\partial_t^2 u(t, x) = \Delta u(t, x) \quad (\text{Wave})$$

$$\left| \begin{array}{ll} \partial_t^2 u(t, x) = \Delta u(t, x) & (t, x) \in I \times \mathbb{R}^d \\ u(0, x) = u_0(x) & x \in \mathbb{R}^d \\ \partial_t u(0, x) = \dot{u}_0(x) & x \in \mathbb{R}^d \end{array} \right. \quad (\text{Free Wave})$$

$$\left| \begin{array}{ll} \partial_t^2 u(t, x) = \Delta u(t, x) & (t, x) \in I \times \Omega \\ u(0, x) = u_0(x) & x \in \Omega \\ \partial_t u(0, x) = \dot{u}_0(x) & \\ u(t, x) = g(t, x) & x \in \partial\Omega \end{array} \right. \quad (4)$$

Intuition:

Symmetries:

Solution:

Conservation Laws:

$$\left| \begin{array}{ll} (\square + m^2) u(t, x) = 0 & (t, x) \in I \times \mathbb{R}^d, \ m \geq 0 \\ u(0, x) = u_0(x) & x \in \mathbb{R}^d \\ \partial_t u(0, x) = \dot{u}_0(x) & \end{array} \right. \quad (\text{Free Klein–Gordon})$$

4.4 Transport Equation

$$\left| \begin{array}{ll} \partial_t u(t, x) + a(t, x) \cdot \nabla u(t, x) = 0 & (t, x) \in I \times \mathbb{R}^d \\ u(0, x) = u_0(x) & x \in \mathbb{R}^d \end{array} \right. \quad (\text{Transport})$$

$$\left| \begin{array}{ll} \partial_t u(t, x) + \nabla \cdot F(u(t, x)) = 0 & (t, x) \in I \times \mathbb{R}^d \\ u(0, x) = u_0(x) & x \in \mathbb{R}^d \end{array} \right. \quad (\text{Conservation})$$

Intuition:

The transport equation describes ...

Solution:

$$\begin{aligned} \partial_t u(t, \gamma(t)) &= \partial_1 u(t, \gamma(t)) + \dot{\gamma}(t) \cdot \nabla_2 u(t, \gamma(t)) \\ \dot{\gamma}(t) &= a(t, \gamma(t)) \end{aligned}$$

Conservation Laws:

The transport equation preserves the L_x^p -norm of the solution for $p \in [1, \infty]$

$$\|u(t, x)\|_{L_x^p(\mathbb{R}^d)} = \|u_0\|_{L_x^p(\mathbb{R}^d)}, \quad t \in I,$$

reflecting the conservation of the transported quantity along the flow.

4.5 Disspersive PDEs

Schrödinger Equation

$$\left| \begin{array}{ll} \partial_t u(t, x) = i\Delta u(t, x) & (t, x) \in I \times \mathbb{R}^d \\ u(0, x) = u_0(x) & x \in \mathbb{R}^d \end{array} \right. \quad (\text{Free Schrödinger})$$

Linear propagator is given by $e^{it\Delta}$

Power nonlinear Schrödinger

$$\left| \begin{array}{ll} \partial_t u = i\partial_x^2 u \pm |u|^{p-1}u & \\ u(0, x) = u_0(x), & x \in \mathbb{R}, t \in \mathbb{R} \end{array} \right. \quad (\text{PNLS})$$

Derivative nonlinear Schrödinger

$$\left| \begin{array}{ll} \partial_t u = i\partial_x^2 u + \lambda \partial_x(|u|^2 u) & \\ u(0, x) = u_0(x), & x \in \mathbb{R}, t \in \mathbb{R} \end{array} \right. \quad (\text{DNLS})$$

KdV Equation

Airy equation, Free KdV

$$\left| \begin{array}{ll} \partial_t u = -\partial_x^3 u & \\ u(0, x) = u_0(x), & x \in \mathbb{R}, t \in \mathbb{R} \end{array} \right. \quad (\text{Airy})$$

Linear propagator is given by $e^{-it\partial_x^3}$

Kortweg-de Vries equation

$$\left| \begin{array}{l} \partial_t u = -\partial_x^3 u - \frac{1}{2} \partial_x^2 u^2 \\ u(0, x) = u_0(x), \end{array} \right. \quad x \in \mathbb{R}, t \in \mathbb{R} \quad (\text{KdV})$$

4.6 Fluid PDEs

Let $\mathbf{u} : I \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ be a velocity field, $\nabla p : I \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ a pressure gradient field

$$\left| \begin{array}{l} \partial_t \mathbf{u}(t, x) + (\mathbf{u}(t, x) \cdot \nabla) \mathbf{u}(t, x) + \nabla p(t, x) = 0 \\ \nabla \cdot \mathbf{u}(t, x) = 0 \\ \mathbf{u}(0, x) = \mathbf{u}_0(x) \end{array} \right. \quad \begin{array}{l} (t, x) \in I \times \mathbb{R}^d, \\ \\ x \in \mathbb{R}^d \end{array} \quad (\text{Incompressible Euler})$$

Let $\nu > 0$ be kinematic viscosity coefficient

$$\left| \begin{array}{l} \partial_t \mathbf{u}(t, x) + (\mathbf{u}(t, x) \cdot \nabla) \mathbf{u}(t, x) + \nabla p(t, x) = \nu \Delta \mathbf{u}(t, x) \\ \nabla \cdot \mathbf{u}(t, x) = 0, \\ \mathbf{u}(0, x) = \mathbf{u}_0(x) \end{array} \right. \quad \begin{array}{l} (t, x) \in I \times \mathbb{R}^d \\ \\ x \in \mathbb{R}^d \end{array} \quad (\text{Incompressible Navier–Stokes})$$

Energy dissipation

$$\partial_t \int |\mathbf{u}| \, dx = -2\nu \int |\nabla \mathbf{u}| \, dx$$

4.7 Solid Mechanics

Small-deformation (linearized) regime. $\mathbf{u} : I \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ is a displacement. The strain tensor is a symmetric gradient (symmetrized Jacobian)

$$e(\mathbf{u}) = \frac{1}{2}(\nabla \mathbf{u} + \nabla \mathbf{u}^\top)$$

Its diagonal entries are normal strains, off-diagonal are shear strains. Let v be a directional vector and h some small length. Then, the relative change of length is encoded in

$$h(e(\mathbf{u})v \mid v)$$

Strain tensor also is invariant under rigid motions of rotations and translations.

Cauchy stress tensor is a matrix for isotropic linear elastic solids, under Hooke's law

$$\sigma(\mathbf{u}) = 2\mu e(\mathbf{u}) + \lambda(\nabla \cdot \mathbf{u})\mathbf{1}_d$$

Let $\rho > 0$ be mass density, $\mu > 0$ Sheer modulus, $\lambda > 0$ first Lamé parameter

$$\left| \begin{array}{ll} \rho \partial_t^2 \mathbf{u}(t, x) - \nabla \cdot \sigma(\mathbf{u}(t, x)) = \mathbf{f}_{\text{ext}}(t, x) & (t, x) \in I \times \mathbb{R}^d \\ \mathbf{u}(0, x) = \mathbf{u}_0(x), \quad \partial_t \mathbf{u}(0, x) = \dot{\mathbf{u}}_0(x) & x \in \mathbb{R}^d \end{array} \right. \quad \text{(Linear Elasticity)}$$

Young's modulus

$$E = \frac{\mu(3\lambda + 2\mu)}{\lambda + \mu}$$

Poisson's coefficient

$$\nu = \frac{\lambda}{2(\lambda + \mu)}$$

5.0 Optimization

5.1 General Notions of Optimization

5.1 Theorem [First, Second Order Conditions]

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ and $x^* \in \mathbb{R}^d$ be a local minimizer of f (unconstrained).

1) if $f \in C^1$, then

$$\nabla f(x^*) = 0$$

2) if $f \in C^2$, then at x^* , then Hessian is *PSD*

$$\mathcal{H}f(x^*) \in \mathcal{S}_+^n$$

5.2 Definition [Convexity]

$K \subset \mathbb{R}^d$ is said to be convex iff

$$(\forall x, y \in K, \forall \theta \in [0, 1]) : \quad \theta x + (1 - \theta)y \in K$$

5.3 Definition [Convexity]

$V \subset \mathbb{R}^d$ is said to be a cone iff

$$(\forall x \in V, \forall t \in \mathbb{R}_+) : \quad tx \in V$$

5.4 Definition [Lower Semi-Continuous Function]

A function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is lower semi-continuous (l.s.c.) iff

$$\liminf_{x \rightarrow x_0} f(x) \geq f(x_0), \quad \forall x_0 \in \mathbb{R}^d$$

Equivalently, iff $\text{epi}(f)$ is closed.

5.5 Definition [Convex Function]

Let $F : \mathbb{R}^d \rightarrow \overline{\mathbb{R}}$ be a function extended by infinity.

1) F is convex iff

$$F(\theta x + (1 - \theta)y) \leq \theta F(x) + (1 - \theta)F(y)$$

2) F is strictly convex iff

$$F(\theta x + (1 - \theta)y) < \theta F(x) + (1 - \theta)F(y)$$

3) F is γ -strongly convex iff

$$F(\theta x + (1 - \theta)y) \leq \theta F(x) + (1 - \theta)F(y) - \frac{\gamma}{2}\theta(1 - \theta) \|x - y\|_2^2$$

5.6 Proposition [Differential Characterization of Convexity]

5.7 Definition [Legendre Dual]

Let $F : \mathbb{R}^d \rightarrow \overline{\mathbb{R}}$ be a function extended by infinity. Its Legendre / Fenchel / Convex dual is the function $F^* : \mathbb{R}^d \rightarrow \overline{\mathbb{R}}$

$$F^*(y) = \sup_{x \in \mathbb{R}^d} (x | y) - F(x)$$

KKT Conditions

5.8 Theorem [Lagrange Multipliers, Equality Constraint]

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ and $\mathbf{g} : \mathbb{R}^d \rightarrow \mathbb{R}^n$ be C^1 . Consider the constrained problem:

$$\left| \begin{array}{l} \arg \min_{x \in \mathbb{R}^d} f(x) \\ \mathbf{g}(x) = 0 \end{array} \right.$$

Define the **Lagrangian**

$$\mathcal{L} : \begin{array}{ccc} \mathbb{R}^d \times \mathbb{R}^n & \longrightarrow & \mathbb{R} \\ (x, \lambda) & \longmapsto & f(x) + (\lambda | \mathbf{g}(x))_{\mathbb{R}^n} \end{array}$$

If

1) $x^\star$ is a local minimizer

2) $\nabla g_1(x_\star), \dots, \nabla g_n(x_\star)$ are linearly independent (equality constraint qualification)

Then, there exists $\lambda^\star \in \mathbb{R}^n$, such that the pair $(x^\star, \lambda^\star)$ satisfies the **Lagrangian stationarity**:

$$\nabla_x \mathcal{L}(x^\star, \lambda^\star) = 0$$

5.9 Theorem [Lagrange Multipliers, Inequality Constraint]

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ and $\mathbf{h} : \mathbb{R}^d \rightarrow \mathbb{R}^m$ be C^1 . Consider the constrained problem:

$$\left| \begin{array}{l} \arg \min_{x \in \mathbb{R}^d} f(x) \\ \mathbf{h}(x) \leq 0 \end{array} \right.$$

Define the **Lagrangian**

$$\mathcal{L} : \begin{array}{ccc} \mathbb{R}^d \times \mathbb{R}_+^m & \longrightarrow & \mathbb{R} \\ (x, \eta) & \longmapsto & f(x) + (\eta \mid \mathbf{h}(x))_{\mathbb{R}^m} \end{array}$$

If

1) $x^\star$ is a local minimizer

2) Inequality constraints qualifications are satisfied

Then, there exists $\lambda^\star \in \mathbb{R}^n$, such that the pair $(x^\star, \lambda^\star)$ satisfies the **Lagrangian stationarity**:

$$\nabla_x \mathcal{L}(x^\star, \lambda^\star) = 0$$

5.10 Theorem [Karush–Kuhn–Tucker (KKT) Conditions]

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ and $\mathbf{g} : \mathbb{R}^d \rightarrow \mathbb{R}^n$ and $\mathbf{h} : \mathbb{R}^d \rightarrow \mathbb{R}^m$ be C^1 . Consider:

$$\left| \begin{array}{l} \arg \min_{x \in \mathbb{R}^d} f(x) \\ \mathbf{g}(x) = 0 \\ \mathbf{h}(x) \leq 0 \end{array} \right.$$

Define the **Lagrangian**

$$\begin{aligned} \mathcal{L} : \quad \mathbb{R}^d \times \mathbb{R}^n \times \mathbb{R}^m &\longrightarrow \mathbb{R} \\ (x, \lambda, \eta) &\longmapsto f(x) + (\lambda \mid \mathbf{g}(x))_{\mathbb{R}^n} + (\eta \mid \mathbf{h}(x))_{\mathbb{R}^m} \end{aligned}$$

If

- 1) $x^\star$ is a local minimizer
- 2) Equality constraint qualifications are satisfied
- 3) Inequality constraint qualifications are satisfied

Then, there exist multipliers $\lambda^\star \in \mathbb{R}^n$ and $\eta^\star \in \mathbb{R}_+^m$ such that:

i) **Stationarity:**

$$\nabla_x \mathcal{L}(x^\star, \lambda^\star, \eta^\star) = 0$$

ii) **Primal feasibility:**

$$\mathbf{g}(x^\star) = 0, \quad \mathbf{h}(x^\star) \leq 0$$

iii) **Complementary slackness:**

$$\eta^\star \odot \mathbf{h}(x^\star) = 0$$

Lagrangian–Hamiltonian Duality

5.11 Definition [Hamiltonian]

Let $\mathcal{L}$ be a Lagrangian as above, define $H : \mathbb{R}^n \times \mathbb{R}_+^m \rightarrow \mathbb{R}$

$$H(\lambda, \eta) := \inf_{x \in \mathbb{R}^d} \mathcal{L}(x, \lambda, \eta)$$

5.12 Theorem [Weak Duality]

It always holds that

$$d^\star \leq p^\star$$

Where,

$$\left| \begin{array}{l} p^* := \min_{x \in \mathbb{R}^d} f(x) \\ \mathbf{g}(x) = 0 \\ \mathbf{h}(x) \leq 0 \end{array} \right| \quad \left| \begin{array}{l} d^* := \sup_{\substack{\lambda \in \mathbb{R}^n \\ \eta \in \mathbb{R}_+^m}} H(\lambda, \eta) \end{array} \right|$$

5.13 Theorem [Lagrangian Saddle Point]

Let $\mathcal{L}$ be a Lagrangian of an optimization problem. Then,

$$d^* = \sup_{\substack{\lambda \in \mathbb{R}^n \\ \eta \in \mathbb{R}_+^m}} \inf_{x \in \mathbb{R}^d} \mathcal{L}(x, \lambda, \eta) \leq p^* = \inf_{x \in \mathbb{R}^d} \sup_{\substack{\lambda \in \mathbb{R}^n \\ \eta \in \mathbb{R}_+^m}} \mathcal{L}(x, \lambda, \eta)$$

And for primal-dual optimizers (x^*, λ^*, η^*) are the saddle points of $\mathcal{L}$

$$\mathcal{L}(x^*, \lambda, \eta) \leq \mathcal{L}(x^*, \lambda^*, \eta^*) \leq \mathcal{L}(x, \lambda^*, \eta^*)$$

5.2 Algorithms

5.14 Proposition [Gradient Descent]

5.15 Proposition [Newton Algorithm]

6.0 Probability

6.1 Basic Definitions and Results

6.1 Definition [Probability Space]

Let $(\Omega, \mathcal{A})$ be a measurable space; P be a probability measure (finite and normalized to 1). Then $(\Omega, \mathcal{A}, P)$ is called a **probability space**.

$$P : \mathcal{A} \rightarrow [0, 1]$$

6.2 Definition [Random Variable]

Let $X : (\Omega, \mathcal{A}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ be measurable, then it's called a real random variable.

6.3 Definition [Law / Distribution of a random variable]

For a random variable X we have

$$\begin{aligned} \mathcal{A} &\xrightarrow{X} \mathcal{B} \\ [0, 1] &\xleftarrow{P} \mathcal{A} \xleftarrow{X^{-1}} \mathcal{B} \end{aligned}$$

Now define

$$\begin{aligned} \mathcal{L}_X : \mathcal{B}(\mathbb{R}) &\rightarrow [0, 1] \\ A &\mapsto \mathcal{L}_X(A) := P \circ X^{-1}(A) = P(X^{-1}(A)) \end{aligned}$$

$\mathcal{L}_X$ is called a **law** (distribution) of a r.v. X

6.4 Definition [Cumulative Distribution Function]

Let $\mathcal{L}_X$ be a law of X

$$\begin{aligned} F_X : \mathbb{R} &\rightarrow [0, 1] \\ x &\mapsto \mathcal{L}_X((-\infty, x]) \end{aligned}$$

F_X is called the **Cumulative Distribution Function**

6.5 Definition [Density]

Let X be a r.v. with law $\mathcal{L}_X$, the **density** of X w.r.t. to a measure ν on $\mathcal{B}(\mathbb{R})$ is a function $f_X : \mathbb{R} \rightarrow \mathbb{R}_+$ such that

$$\mathcal{L}_X(A) = \int_A f_X d\nu, \quad \forall A \in \mathcal{B}(\mathbb{R})$$

If $\nu = \text{Leb}_{\mathbb{R}}$

$$\mathcal{L}_X(A) = \int_A f_X(t) dt, \quad \forall A \in \mathcal{B}(\mathbb{R})$$

6.6 Definition [Atom of a Random Variable]

Let X be a r.v., an element $a \in \mathbb{R}$ is called an atom of X if

$$P(X = a) \neq 0$$

6.7 Definition [Expectation]

Let $X \in L^1(\Omega : \mathbb{R}; P)$, then

$$\mathbb{E}[X] := \int_{\Omega} X dP = \int_{\Omega} X(\omega) dP(\omega)$$

Furthermore, by **Transfer Theorem** $g : \mathbb{R} \rightarrow \mathbb{R}$

$$\mathbb{E}[g(X)] = \int_{\Omega} g(X(\omega)) dP(\omega) = \int_{\mathbb{R}} g(t) d\mathcal{L}_X(t)$$

6.8 Definition [Variance]

Let $X \in L^2(\Omega : \mathbb{R}; P)$

$$\begin{aligned} \text{Var}[X] &= \mathbb{E}[X^2] - \mathbb{E}[X]^2 \\ &= \mathbb{E}[(X - \mathbb{E}[X])^2] \end{aligned}$$

6.9 Definition [Covariance]

Let $X, Y \in L^2(\Omega : \mathbb{R}; P)$

$$\begin{aligned} \text{cov}[X, Y] &= \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] \\ &= \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] \end{aligned}$$

6.10 Definition [Covariance Matrix]

Let $\mathbf{X} = (X_1, \dots, X_d)^\top$ be a random vector in $L^2(\Omega : \mathbb{R}^d)$, its variance is defined as its covariance matrix $\mathbf{\Sigma} \in \mathbb{R}^{d \times d}$

$$\text{Var}[\mathbf{X}] := \mathbf{\Sigma} := \begin{bmatrix} \text{Var}[X_1] & \cdots & \text{cov}[X_1, X_d] \\ \vdots & \ddots & \vdots \\ \text{cov}[X_d, X_1] & \cdots & \text{Var}[X_d] \end{bmatrix} = (\text{cov}[X_i, X_j])_{i,j}$$

Covariance matrix is always in $\mathcal{S}_+^d$

6.11 Remark

If X takes values in $\mathbb{Z}_+$,

$$\begin{aligned}\mathbb{E}[X] &= \sum_{k \geq 0} k P(X = k) \\ \mathbb{E}[X] &= \sum_{k \geq 1} P(X \geq k)\end{aligned}$$

6.12 Definition [Convergence almost surely]

$(X_n)_{n \in \mathbb{N}}$ converges to X almost surely whenever

$$\begin{aligned}X_n &\xrightarrow[n \rightarrow \infty]{\text{a.s.}} X \\ P\left(\left\{\omega : \lim_{n \rightarrow \infty} X(\omega) = X_n(\omega)\right\}\right) &= 1\end{aligned}$$

6.13 Definition [Convergence in Probability]

$(X_n)_{n \in \mathbb{N}}$ converges to X in Probability whenever

$$\begin{aligned}X_n &\xrightarrow[n \rightarrow \infty]{P} X \\ P(\|X(\omega) - X_n(\omega)\| > \epsilon) &= 0\end{aligned}$$

6.14 Definition [Convergence in Distribution]

$(X_n)_{n \in \mathbb{N}}$ converges to X in Distribution whenever for all h bounded, continuous

$$\begin{aligned}X_n &\xrightarrow[n \rightarrow \infty]{(d)} X \\ \mathbb{E}[h(X_n)] &\xrightarrow[n \rightarrow \infty]{} \mathbb{E}[h(X)]\end{aligned}$$

6.15 Theorem [Borel-Canteli]

Let $(E_n)_n$ be a sequence of events

$$\sum_{n \in \mathbb{N}} P(E_n) < \infty \implies P(\limsup_{n \rightarrow \infty} E_n) = P\left(\bigcap_{n=1} \bigcup_{k=n} E_k\right) = 0.$$

Additionally, if E_n are independent,

$$\sum_{n \in \mathbb{N}} P(E_n) = \infty \implies P(\limsup_{n \rightarrow \infty} E_n) = P\left(\bigcap_{n=1} \bigcup_{k=n} E_k\right) = 1.$$

6.16 Definition [Characteristic Function]

Let X be a r.v. in $\mathbb{R}$

$$\begin{aligned} \varphi_X : \mathbb{R} &\rightarrow \mathbb{C} \\ t &\mapsto \mathbb{E}[\exp(itX)] = \mathbb{E}[e^{itX}] \end{aligned}$$

Let $\mathbf{X}$ be a r.v. in $\mathbb{R}^d$

$$\begin{aligned} \varphi_{\mathbf{X}} : \mathbb{R}^d &\rightarrow \mathbb{C} \\ t &\mapsto \mathbb{E}[\exp(it \cdot \mathbf{X})] = \mathbb{E}[e^{it^\top \mathbf{X}}] \end{aligned}$$

6.17 Proposition [Relationship Between Convergences]

Let $(X_n)_{n \in \mathbb{N}}$ be a sequence of r.v. in $\mathbb{R}$ and X be an r.v. in $\mathbb{R}$

$$\begin{aligned} a.s. &\implies P \implies (d) \\ L^p &\implies L^1 \implies P \end{aligned}$$

Additionally, if $X_n \xrightarrow[n \rightarrow \infty]{(d)} X = c$, where $c \in \mathbb{R}$, then

$$P \iff (d)$$

6.18 Theorem [Portmanteau]

Let $(X_n)_{n \in \mathbb{N}}$ be a sequence of r.v. in $\mathbb{R}$ and X be an r.v. in $\mathbb{R}$.

Let A be a set of atoms of X . Then,

$$X_n \xrightarrow[n \rightarrow \infty]{(d)} X \iff P(X_n \leq t) \xrightarrow[n \rightarrow \infty]{} P(X \leq t), \quad \forall t \in \mathbb{R}/A$$

6.19 Theorem [Continuous Mapping]

Let $(\mathbf{X}_n)_{n \in \mathbb{N}}$ be a sequence of r.v. in $\mathbb{R}^d$ and $\mathbf{X}$ be an r.v. in $\mathbb{R}^d$

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}^p$ be a continuous function.

Then,

$$\begin{aligned} \mathbf{X}_n \xrightarrow[n \rightarrow \infty]{a.s.} \mathbf{X} &\implies f(\mathbf{X}_n) \xrightarrow[n \rightarrow \infty]{a.s.} f(\mathbf{X}) \\ \mathbf{X}_n \xrightarrow[n \rightarrow \infty]{P} \mathbf{X} &\implies f(\mathbf{X}_n) \xrightarrow[n \rightarrow \infty]{P} f(\mathbf{X}) \\ \mathbf{X}_n \xrightarrow[n \rightarrow \infty]{(d)} \mathbf{X} &\implies f(\mathbf{X}_n) \xrightarrow[n \rightarrow \infty]{(d)} f(\mathbf{X}) \end{aligned}$$

6.20 Theorem [Lévy]

Let $(X_n)_{n \in \mathbb{N}}$ be a sequence of r.v. in $\mathbb{R}$ and X be an r.v. in $\mathbb{R}$

Let $\varphi_{X_n}(t)$ denote the characteristic function of X_n .

If

$$\varphi_{X_n}(t) \xrightarrow[n \rightarrow \infty]{} \varphi(t), \quad \forall t \in \mathbb{R}$$

Then,

$$\begin{aligned} \varphi(t) &= \varphi_X(t) \in C^0 \\ X_n &\xrightarrow[n \rightarrow \infty]{(d)} X \end{aligned}$$

6.21 Theorem [Slutsky]

Let $(\mathbf{X}_n)_{n \in \mathbb{N}}$, $(\mathbf{Y}_n)_{n \in \mathbb{N}}$ be sequences of r.v. in $\mathbb{R}^d$, $\mathbb{R}^p$.

Let $\mathbf{X}$ be a r.v. in $\mathbb{R}^d$, let $\mathbf{c} \in \mathbb{R}^p$ be a fixed vector.

Then,

$$\begin{pmatrix} \mathbf{X}_n \\ \mathbf{Y}_n \end{pmatrix} \xrightarrow[n \rightarrow \infty]{(d)} \begin{pmatrix} \mathbf{X} \\ \mathbf{c} \end{pmatrix} \iff \begin{cases} \mathbf{X}_n \xrightarrow[n \rightarrow \infty]{(d)} \mathbf{X} \\ \mathbf{Y}_n \xrightarrow[n \rightarrow \infty]{(d)} \mathbf{c} \end{cases}$$

Theorem gives ($\iff$).

6.22 Proposition [Convergence in Distribution]

Let $(\mathbf{X}_n)_{n \in \mathbb{N}}$, $(\mathbf{Y}_n)_{n \in \mathbb{N}}$ be sequences of r.v. in $\mathbb{R}^d$, $\mathbb{R}^p$.

Let $\mathbf{X}$ be a r.v. in $\mathbb{R}^d$, let $\mathbf{Y} \in \mathbb{R}^p$.

If $\mathbf{X}$ is independent of $\mathbf{Y}$ and $\mathbf{X}_n$ is independent of $\mathbf{Y}_n$ for all $n \in \mathbb{N}$. Then,

$$\begin{pmatrix} \mathbf{X}_n \\ \mathbf{Y}_n \end{pmatrix} \xrightarrow[n \rightarrow \infty]{(d)} \begin{pmatrix} \mathbf{X} \\ \mathbf{Y} \end{pmatrix} \iff \begin{cases} \mathbf{X}_n \xrightarrow[n \rightarrow \infty]{(d)} \mathbf{X} \\ \mathbf{Y}_n \xrightarrow[n \rightarrow \infty]{(d)} \mathbf{Y} \end{cases}$$

6.23 Theorem [R.v. Generator]

Let Z be a real r.v. with density w.r.t. $\text{Leb}_{\mathbb{R}}$. Let F be CDF of Z . Then, $F(Z) \sim \mathcal{U}(0, 1)$

6.2 Limit Theorems

6.24 Theorem [Law of Large Numbers]

Let $X_1, \dots, X_n, \dots$ be i.i.d. random variables with 1 moment; let

$$\bar{X}_n := \frac{1}{n} \sum_{k=1}^n X_k$$

Then,

$$\bar{X}_n \xrightarrow[n \rightarrow \infty]{\text{a.s.}} \mathbb{E}[X_1]$$

6.25 Theorem [Central Limit Theorem]

Let $X_1, \dots, X_n, \dots$ be i.i.d. random variables with 2 moments.

Then,

$$\sqrt{n}(\bar{X}_n - \mathbb{E}[X_1]) \xrightarrow[n \rightarrow \infty]{(d)} \mathcal{N}(0, \text{Var}[X_1])$$

6.26 Theorem [Multivariate Central Limit Theorem]

Let $\mathbf{X}_1, \dots, \mathbf{X}_n, \dots$ be i.i.d. random variables in $\mathbb{R}^d$ with 2 moments.

Then,

$$\sqrt{n}(\bar{\mathbf{X}}_n - \mathbb{E}[\mathbf{X}_1]) \xrightarrow[n \rightarrow \infty]{(d)} \mathcal{N}_d(\mathbf{0}, \Sigma)$$

6.27 Theorem [Delta-Method]

Let $(T_n)_{n \geq 1}$ be a sequence of r.v. with

$$\rho_n(T_n - \theta) \xrightarrow[n \rightarrow \infty]{(d)} \mathcal{N}(0, \sigma^2)$$

for some $\theta \in \mathbb{R}, \sigma^2 > 0, \rho_n$ increasing sequence.

We would say that T_n is asymptotically normal around θ with rate ρ_n and asymptotic variance σ^2 (though it may have nothing to do with the actual limiting variance of T_n).

Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be differentiable at θ with $g'(\theta) \neq 0$. Then,

$$\rho_n(g(T_n) - g(\theta)) \xrightarrow[n \rightarrow \infty]{(d)} \mathcal{N}(0, \sigma^2 \cdot (g'(\theta))^2)$$

In practice, usually, $T_n = \bar{X}_n, \theta = \mathbb{E}[X_1], \sigma^2 = \text{Var}[X_1], \rho_n = \sqrt{n}$.

6.28 Theorem [Multivariate Delta-Method]

Let $(\mathbf{T}_n)_{n \geq 1}$ be a sequence of r.v. in $\mathbb{R}^d$ with

$$\rho_n(\mathbf{T}_n - \boldsymbol{\theta}) \xrightarrow[n \rightarrow \infty]{(d)} \mathcal{N}_d(\mathbf{0}, \boldsymbol{\Sigma})$$

for some $\boldsymbol{\theta} \in \mathbb{R}^d, \boldsymbol{\Sigma} \in \mathcal{S}_+^d$

Let $g : \mathbb{R}^d \rightarrow \mathbb{R}^p$ be differentiable at $\boldsymbol{\theta}$ with $\mathcal{J}_g(\boldsymbol{\theta}) \neq 0$. Then,

$$\rho_n(g(\mathbf{T}_n) - g(\boldsymbol{\theta})) \xrightarrow[n \rightarrow \infty]{(d)} \mathcal{N}_d(\mathbf{0}, \mathcal{J}_g(\boldsymbol{\theta}) \boldsymbol{\Sigma} \mathcal{J}_g^\top(\boldsymbol{\theta})),$$

where $\mathcal{J}_g(\boldsymbol{\theta}) = [\mathcal{J}g](\boldsymbol{\theta})$ is Jacobian of g evaluated at $\boldsymbol{\theta}$

$$\mathcal{J}_g(\mathbf{x}) := \begin{pmatrix} \partial_{x_1} g_1(\mathbf{x}) & \cdots & \partial_{x_d} g_1(\mathbf{x}) \\ \vdots & \ddots & \vdots \\ \partial_{x_1} g_p(\mathbf{x}) & \cdots & \partial_{x_d} g_p(\mathbf{x}) \end{pmatrix} = (\partial_{x_j} g_i(\mathbf{x}))_{i,j} \in \mathbb{R}^{p \times d}$$

6.3 Gaussian and Chi-squared

6.29 Definition [Gaussian (Normal) Distribution]

Univariate Gaussian Distribution $\mathcal{N}(\mu, \sigma^2)$ for some $\mu \in \mathbb{R}, \sigma^2 > 0$ is a distribution on

$\mathcal{B}(\mathbb{R})$ with the following density $f : \mathbb{R} \rightarrow \mathbb{R}_+$, w.r.t. $\text{Leb}_{\mathbb{R}}$

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2} \frac{(x - \mu)^2}{\sigma^2}\right)$$

6.30 Proposition

6.31 Definition [Gaussian Vector]

Let $\mathbf{X} = (X_1, \dots, X_d)^\top$ be a random vector. It is called a Gaussian vector if any linear combination of $X_1, \dots, X_d$ is a univariate Gaussian r.v.

6.32 Proposition

Let $X_1, \dots, X_d$ be independent Gaussian random variables, then $\mathbf{X} = (X_1, \dots, X_d)^\top$ is a Gaussian vector

6.33 Proposition [Multivariate Gaussian Distribution]

Let $\mathbf{X}$ be a Gaussian vector. It has 2 moments, and we can denote $\boldsymbol{\mu} = \mathbb{E}[\mathbf{X}]$, $\boldsymbol{\Sigma} = \text{Var}[\mathbf{X}]$. Then, its characteristic function is

$$\varphi_{\mathbf{X}}(\mathbf{t}) = \mathbb{E}[e^{i\mathbf{t}^\top \mathbf{X}}] = \exp(i\mathbf{t}^\top \boldsymbol{\mu} - \frac{1}{2} \mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t})$$

and the distribution of X can be denoæted as $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$

6.34 Proposition

Let $X \sim \mathcal{N}_d(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, for some $\boldsymbol{\mu} \in \mathbb{R}^d$, $\boldsymbol{\Sigma} \in \mathcal{S}_+^d$.

The following holds

$$X \text{ has a density w.r.t. } \text{Leb}_{\mathbb{R}^d} \iff \boldsymbol{\Sigma} \in \mathcal{S}_{++}^d$$

and the density $f : \mathbb{R}^d \rightarrow \mathbb{R}_+$ is

$$f(\mathbf{x}) = (2\pi)^{-d/2} \det(\mathbf{\Sigma})^{-1/2} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \mathbf{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right)$$

6.35 Proposition

Let $\mathbf{X}$ be a Gaussian vector of size $d \geq 2$. Write

$$X = \begin{pmatrix} Y \\ Z \end{pmatrix} \quad \text{Var}[X] = \begin{bmatrix} \text{Var}[Y] & \text{cov}[Y, Z] \\ \text{cov}[Z, Y] & \text{Var}[Z] \end{bmatrix}$$

The following holds

$$Y \text{ is independent of } Z \iff \text{cov}[Y, Z] = 0$$

6.36 Definition [Chi-squared]

Let $Z_1, \dots, Z_d \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$, then the distribution of the random variable

$$\chi_d^2 := \sum_{i=1}^d Z_i^2$$

is called "chi-squared distribution with d degrees of freedom".

6.37 Theorem [Cochran]

Let $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2)$; $\mu \in \mathbb{R}, \sigma^2 > 0, n \geq 2$.

Denote $\hat{\mu}_n := \bar{X}_n$, $\hat{\sigma}_n^2 := \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2$.

Then,

- 1) $\hat{\mu}_n \sim \mathcal{N}(\mu, \sigma^2/n)$
- 2) $n\hat{\sigma}_n^2/\sigma^2 \sim \chi_{n-1}^2$
- 3) $\hat{\mu}_n$ is independent of $\hat{\sigma}_n^2$

6.38 Definition [Student-t distribution]

Let $d \geq 1$; $U \sim \mathcal{N}(0, 1)$, $V \sim \chi_d^2$ and U independent of V .

Then, the Student-t distribution with d -degrees of freedom is the distribution of

$$T = \frac{U}{\sqrt{V/d}}$$

$T \sim t_d$.

6.39 Example

Let $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2)$, $\mu \in \mathbb{R}$, $\sigma^2 > 0$, $n \geq 2$; $\hat{\sigma}_n := \sqrt{\hat{\sigma}_n^2}$. Then,

$$\sqrt{n-1} \frac{\bar{X}_n - \mu}{\hat{\sigma}_n} \sim t_{n-1}$$

6.4 Matringales

6.40 Definition [Conditional Expectation]

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, $X \in L^1(\Omega)$ a random variable, and $\mathcal{G} \subset \mathcal{F}$ a sub- σ -algebra. The **conditional expectation** of X given $\mathcal{G}$, denoted $\mathbb{E}[X \mid \mathcal{G}]$, is the unique (up to a.s. equality) $\mathcal{G}$ -measurable random variable satisfying

$$\int_G \mathbb{E}[X \mid \mathcal{G}] d\mathbb{P} = \int_G X d\mathbb{P}, \quad \forall G \in \mathcal{G}.$$

6.41 Definition [Martingale]

Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$ be a filtered probability space. A stochastic process $(M_t)_{t \geq 0}$ adapted to $(\mathcal{F}_t)$ is a **martingale** if

$$\mathbb{E}[|M_t|] < \infty, \quad \text{and} \quad \mathbb{E}[M_t \mid \mathcal{F}_s] = M_s \quad \text{a.s. for all } 0 \leq s \leq t.$$

6.5 Stochastic Calculus

6.42 Definition [Brownian Motion]